

Genomics data analysis 基因组学数据分析

分类数据(Categorical data)与富集

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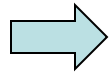
贺圆圆 <heyuanyuan@stu.pku.edu.cn>



课程网页

联系助教加入课程微信群

本节内容



- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

相关性提示因果关系

- 燕子低飞要落雨。
- 日落西北满天红，不是雨来就是风。
- 立秋有雨样样有，立秋无雨收半秋。
- 要知明天热不热，就看夜星密不密。

分类数据

- 分类数据(categorical data)是按照现象的某种属性对其进行分类或分组而得到的反映事物类型的数据
 - 例如，按照性别将人口分为男、女两类；按照经济性质将企业分为国有、集体、私营、其他经济等。“男”、“女”，“国有”、“集体”、“私营”和“其他经济”就是分类数据。
 - 为了便于计算机处理，通常用数字代码来表述各个类别，比如，用1表示“男性”，0表示“女性”，但是1和0等只是数据的代码，它们之间没有数量上的关系和差异。
 - 对应于R语言的factor数据类型

朱喜安．统计学：湖北科学技术出版社，2014

<https://baike.baidu.com/item/%E5%88%86%E7%B1%BB%E6%95%B0%E6%8D%AE>

R factor数据类型

```
mons = c("March", "April", "January", "November", "January",  
  "September", "October", "September", "November", "August",  
  "January", "November", "November", "February", "May", "August",  
  "July", "December", "August", "August", "September", "November",  
  "February", "April")
```

```
mons
```

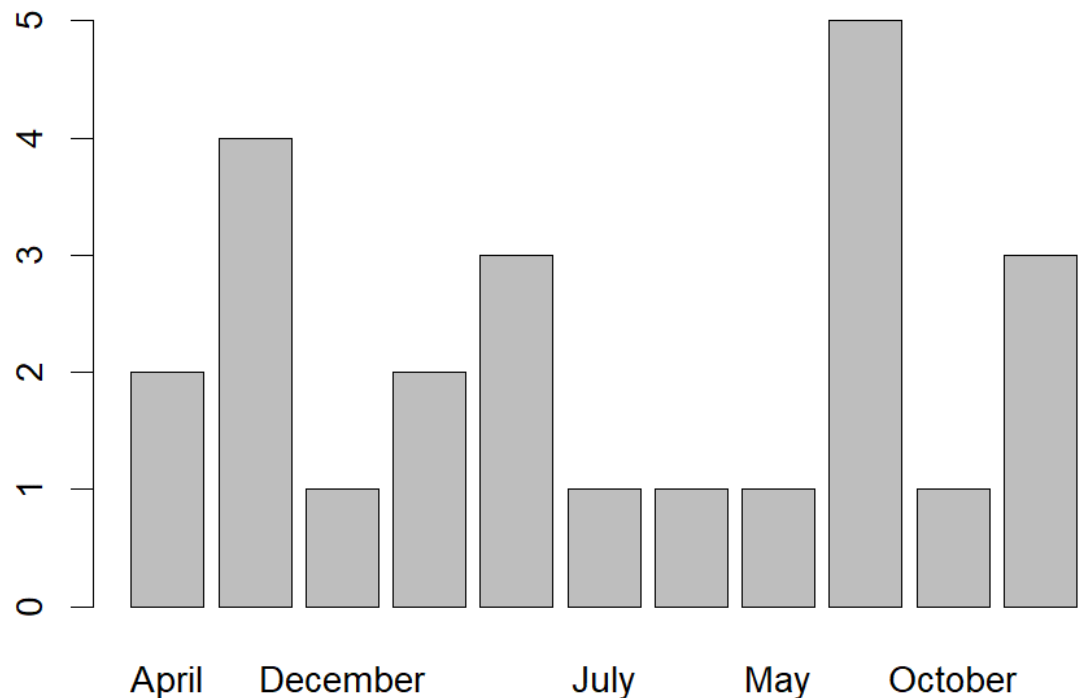
```
mons = factor(mons)
```

```
mons
```

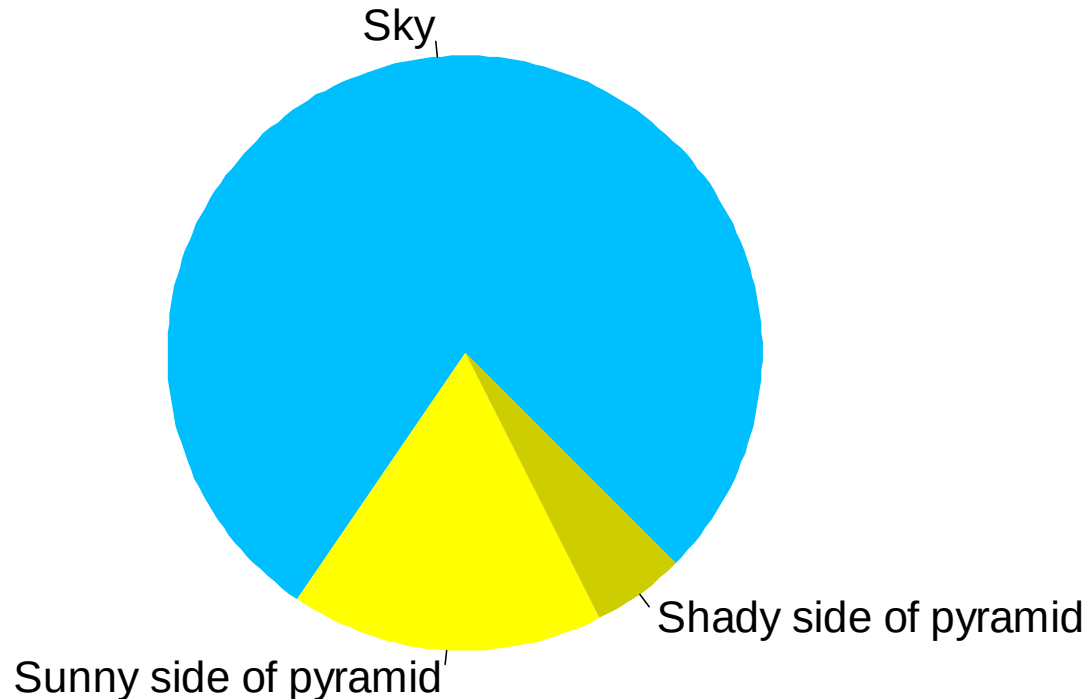
```
table(mons)
```

```
plot(mons)
```

调用对不同数据类型
的缺省plot
函数

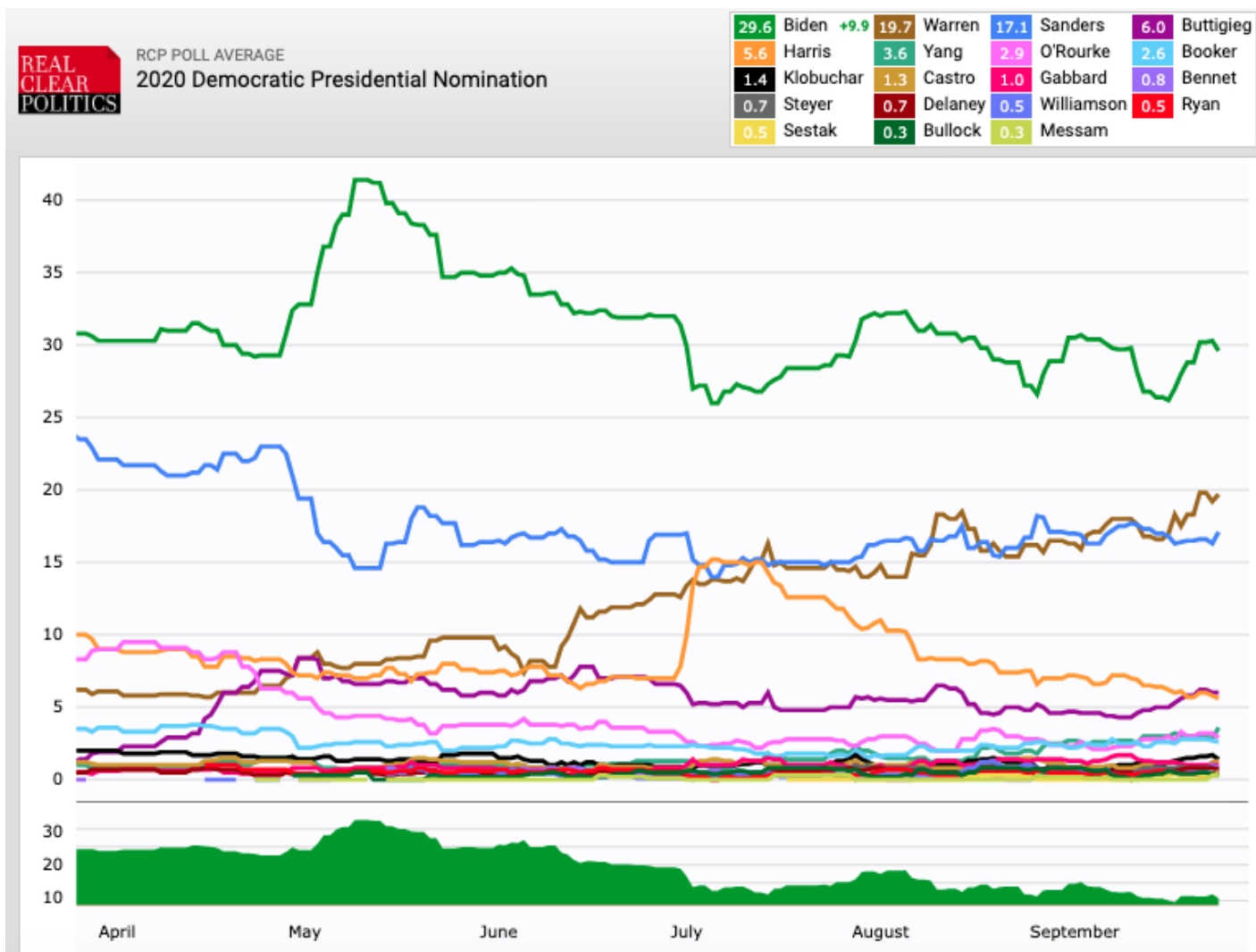


分类数据的常见作图：饼图

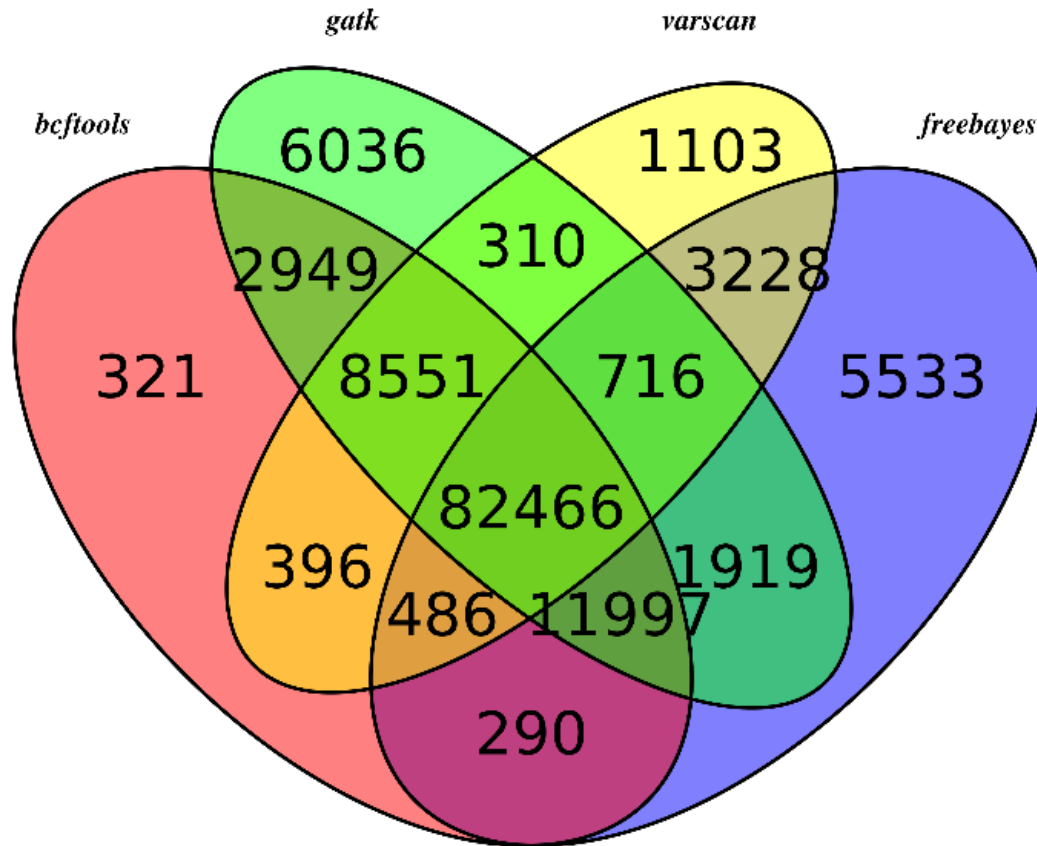


```
pie(c(Sky = 78, "Sunny side of pyramid" = 17,  
      "Shady side of pyramid" = 5), init.angle = 315,  
    col = c("deepskyblue", "yellow", "yellow3"),  
    border = FALSE)
```

分类数据常转化为比例做比较



韦恩图 (Venn Diagram)



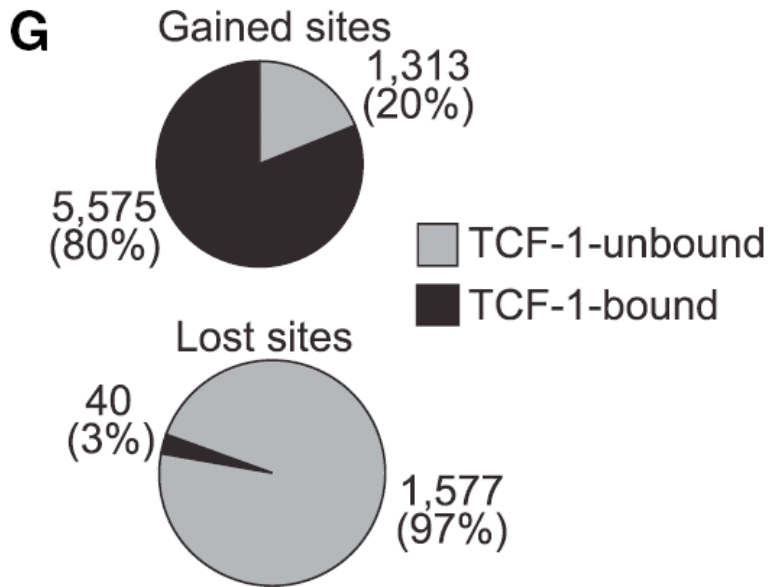
找两个、三个集合的共同元素？

用R语言绘制韦恩图：<https://www.plob.org/article/7347.html>

```
install.packages("VennDiagram")
```


文献举例：比例的比较

生信思路：相关性提示因果关系：TCF-1结合的位置，染色质开放性增强较多



Cell 17 YY1 Is a Structural Regulator of Enhancer-Promoter Loops

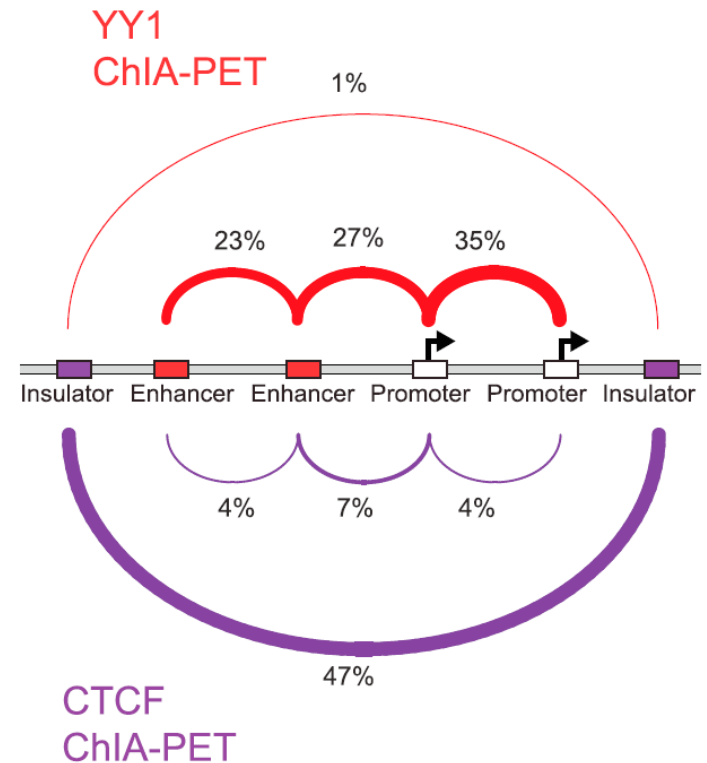


Figure 4. [TCF-1 Can Bind to Nucleosomes and Create Chromatin Accessibility in Fibroblasts](#)
(G) TCF-1 bound to 5,575 (80%) gained accessible sites in contrast to only 40 (3%) lost sites.

Immunity 18 Lineage-Determining Transcription Factor TCF-1 Initiates the Epigenetic Identity of T Cells

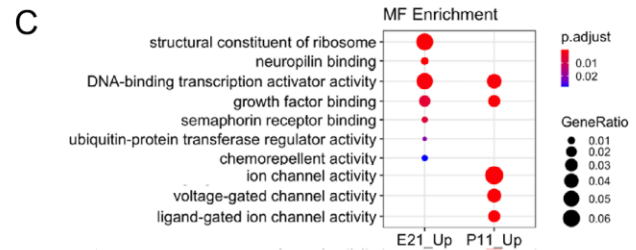
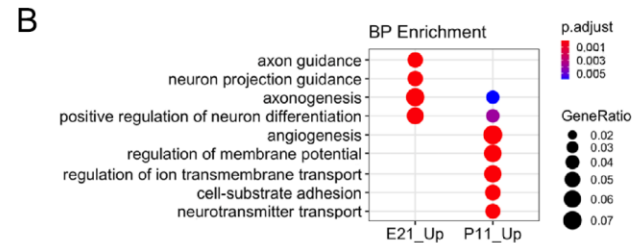
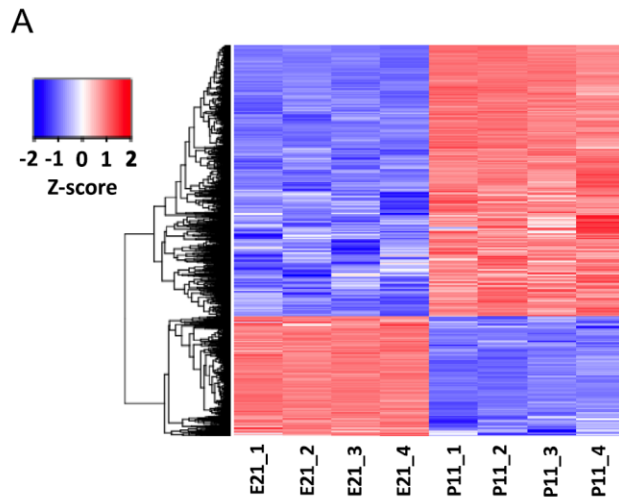
文献：通过染色质开放性研究视神经再生

- **实验设计**：To uncover mechanisms that promote regeneration of RGC (Retinal Ganglion Cells) axons, we identified **transcription factors (TF) and open chromatin regions** that are enriched in rat embryonic RGCs (high axon growth capacity) compared to postnatal RGCs (low axon growth capacity).
- **组学分析**：Binding motifs for TFs such as CREB, CTCF, JUN and YY1 were enriched in the regions of the chromatin that were more accessible in embryonic RGCs.
- **功能验证**：Overexpression of CREB fused to the VP64 transactivation domain in mouse RGCs promoted axon regeneration after optic nerve injury.
- **公共数据**：The raw FASTQ files at the NCBI GEO database: **GSE163564**

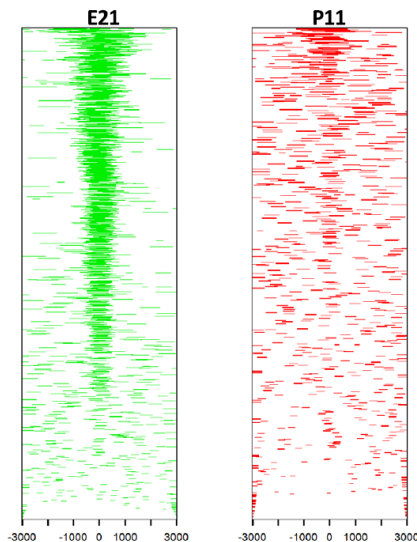
生信思路：比较阅读不同影响力的文献，思考实验设计、数据分析、生物发现和意义方面的异同

SR 21 Genome-wide chromatin accessibility analyses provide a map for enhancing optic nerve regeneration

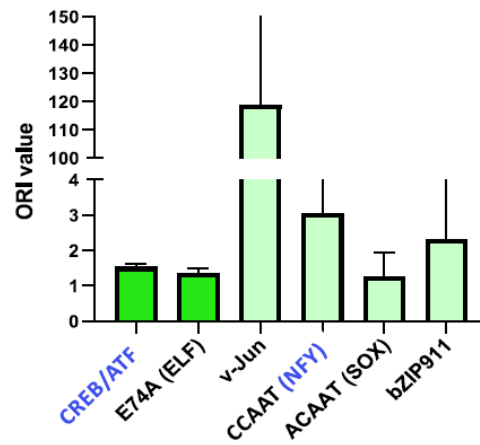
文献：通过染色质开放性研究视神经再生



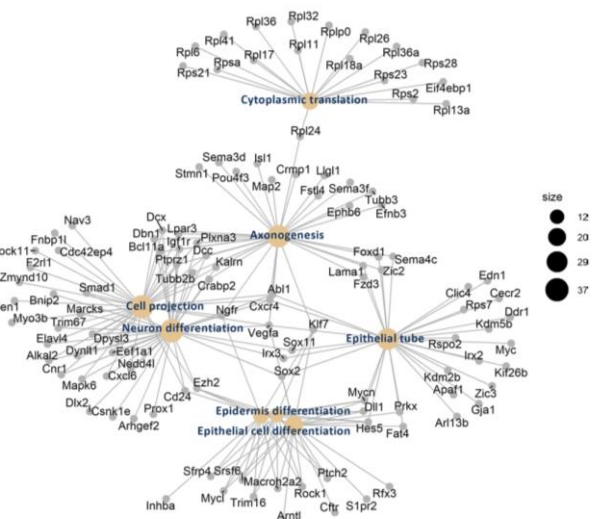
DORs distribution around TSS



E21 enriched motifs (ORI)



B GO Biological Process

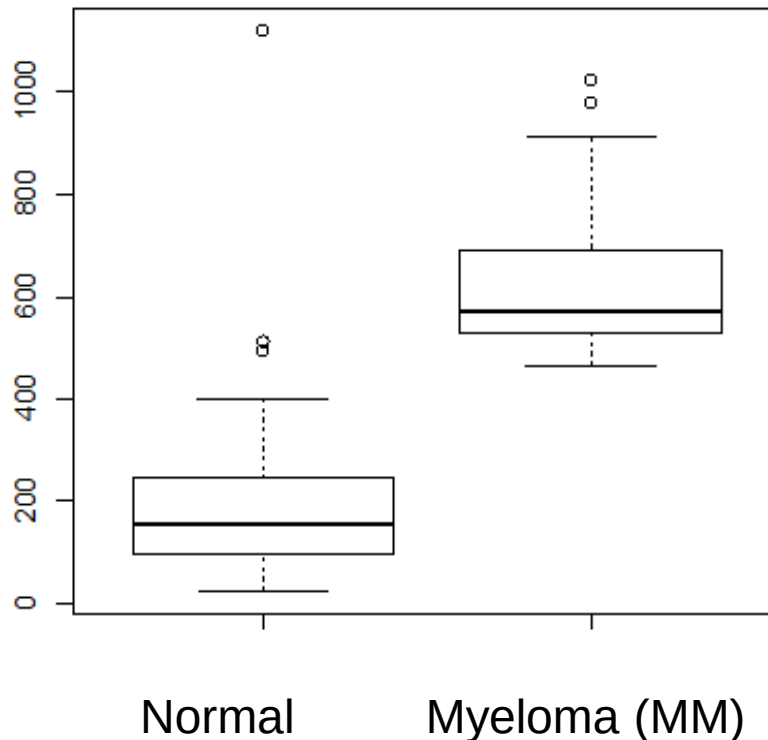


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Differentially expressed genes

- What genes are differentially expressed between the normal and myeloma (MM) groups?



t-test

Test the **null hypothesis** that Normal and MM values come from the distributions with the same mean. (**no expression difference between the two groups**)

Two-sample t-test: question

- Data points are from two groups, $x_{11}, \dots, x_{1,n_1}$ and $x_{21}, \dots, x_{2,n_2}$, which we assume are sampled independently from normal distributions:

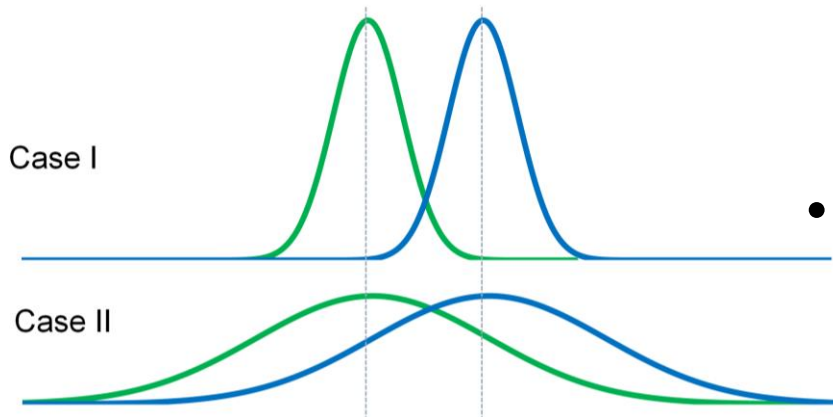
$$N(\mu_1, \sigma_1^2) \text{ and } N(\mu_2, \sigma_2^2).$$

- Question: to test the null hypothesis (H_0):

$$\mu_1 = \mu_2$$

统计检验要素:

- 选择零假设和备选假设
- 如何设计统计量
- 统计量是什么分布



- Devise a statistic, which is big when H_1 is true ($\mu_1 \neq \mu_2$)
- `t.test()` function in R

Equal variance case

- Assumes the two groups have the **equal variance**
 - First calculate a pooled standard deviation (**SD**; $S_{x1,x2}$) based on the sample SD from the two groups

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

对 X_i 的标准差
 σ 的估计值

$$S_{X_1X_2} = \sqrt{\frac{(n_1 - 1)S_{X_1}^2 + (n_2 - 1)S_{X_2}^2}{n_1 + n_2 - 2}}$$

- Use it to compute Standard Error of Difference of Mean (**SEDM**)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_{X_1X_2} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

对分子的标准
差的估计值

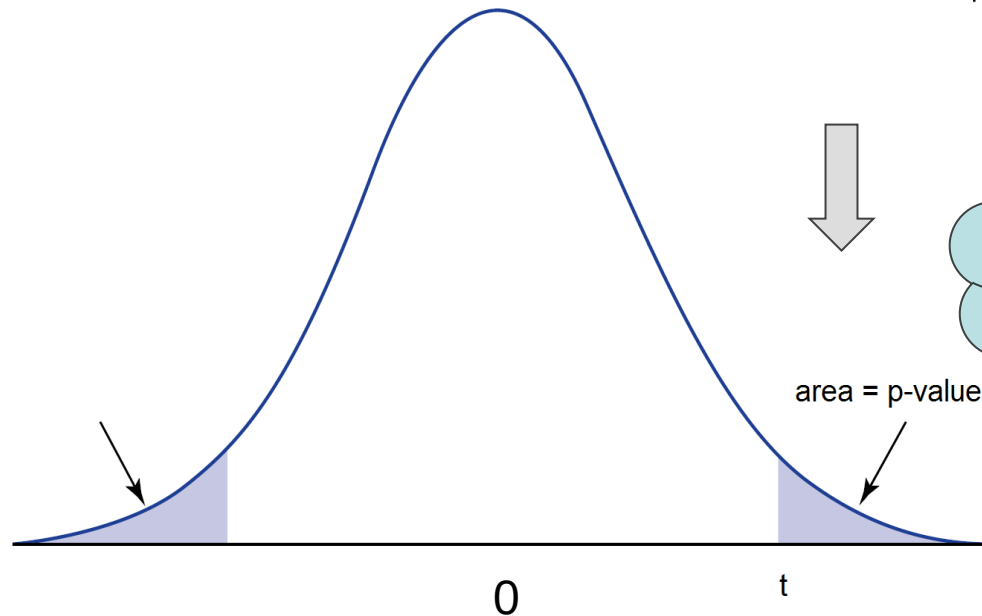
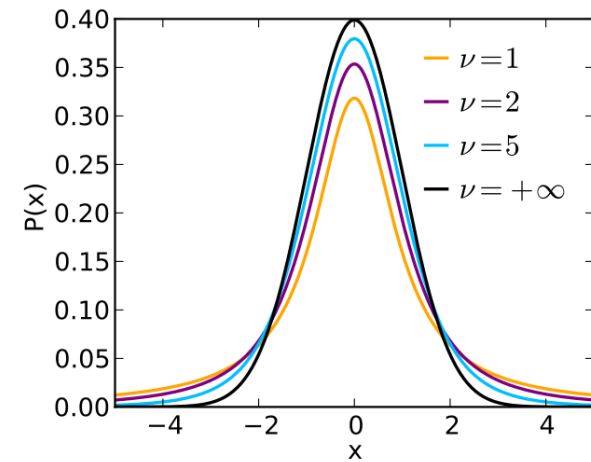
- The t-statistic follows a **t distribution** with $n_1 + n_2 - 2$ degrees of freedom (d.f.)

计算标准差时
估计了2个均值
，独立数据点
减少了2个

From t-statistic to p-value

$$t = \frac{\bar{x}_2 - \bar{x}_1}{\text{SEDM}}$$

设计这样的统计量 t ，以便推导出 t 的理论分布



P value: 零假设下观测到极端值的概率

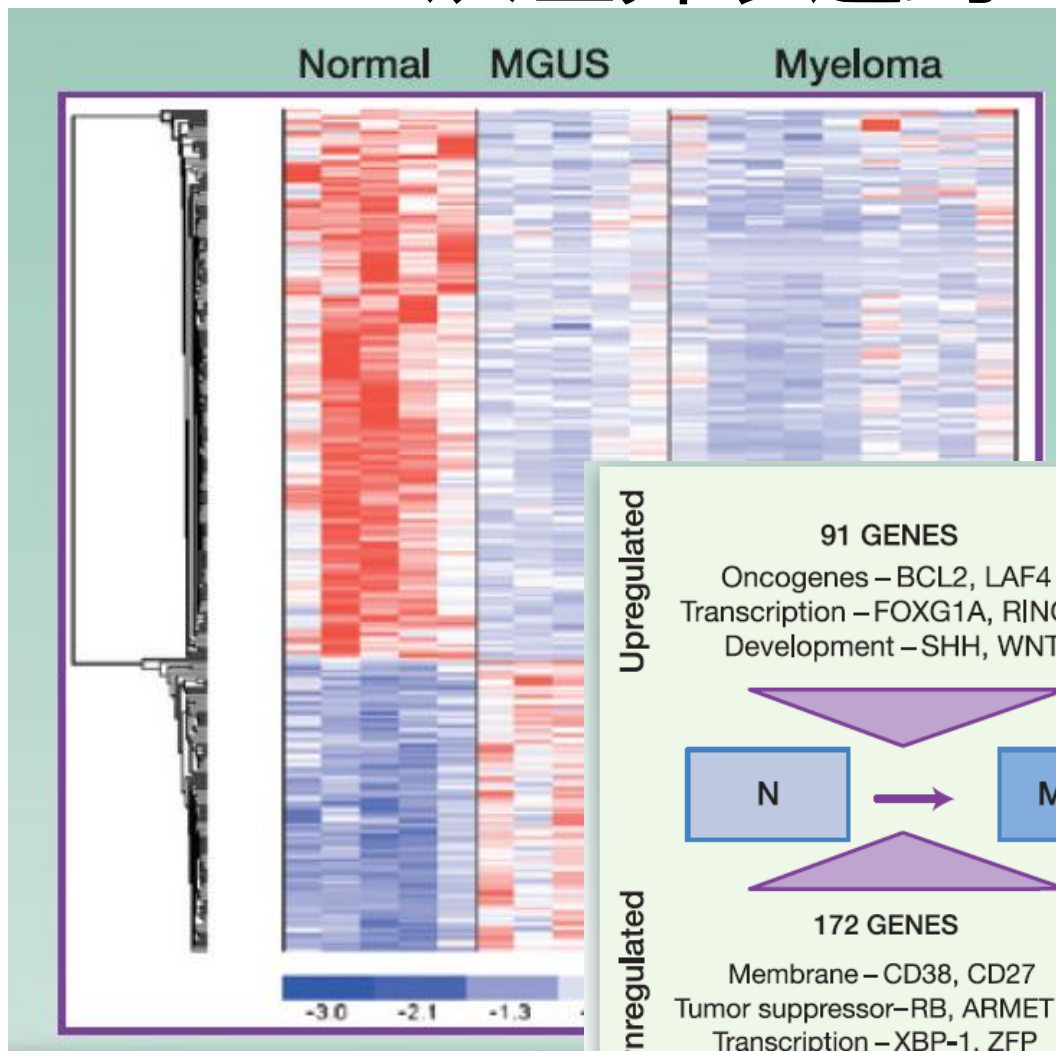
The distribution of the t-statistic under the null hypothesis: $\mu_1 = \mu_2$

A myeloma expression data set

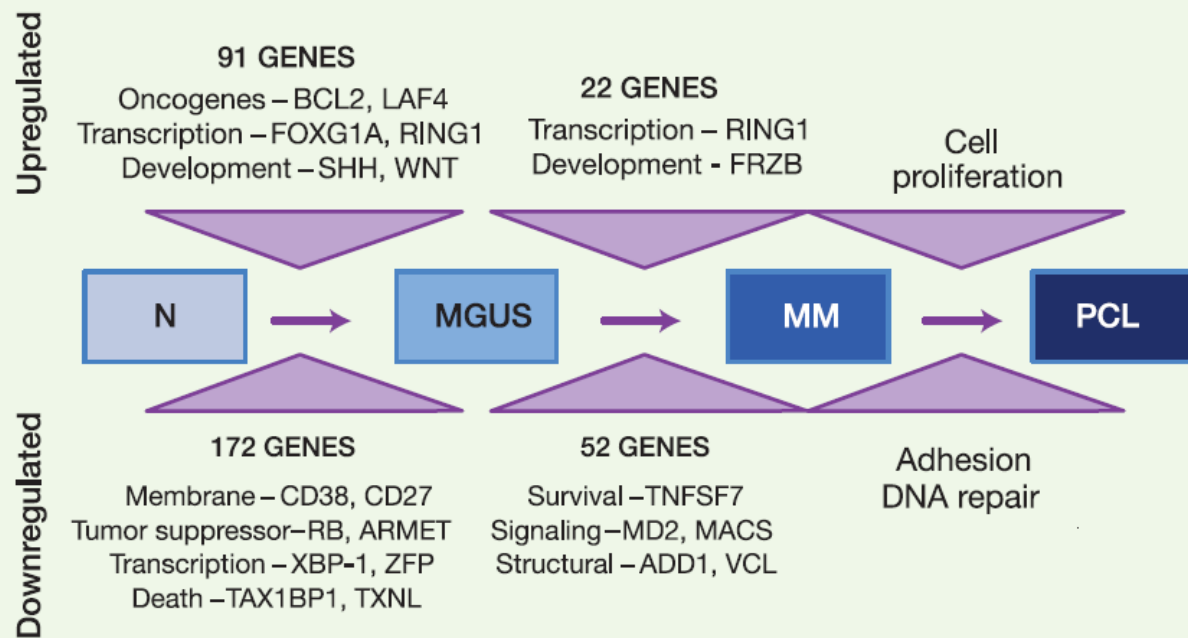
5 normal sample and 9 myeloma (MM) samples 12558 genes (rows)

probe set	gene	Normal m4	Normal m4	Normal m4	Normal m4	Normal m4	MM m282	MM m331a	MM m332a	MM m333a	MM m334a	MM m353a	MM m408a	MM m423a
31307_at	pre-T/NK c	28.53	32.61	29.56	36.55	33.19	25.1	32.79	34.3	35.44	28.48	29.55	22.28	28.77
31308_at	pre-T/NK c	69.14	53.69	52.78	62.07	58.74	67.88	85.82	83.54	85.91	60.93	62.82	47.17	77.07
31309_r_a	Human bre	16.9	67.7	27.61	46.16	51.46	45.62	35.57	32.62	35.14	96.18	45.94	63.2	38.27
31310_at	glycine rec	67.42	49.55	55.51	59.57	68.42	91.06	91.23	83.66	76.37	71.23	74.95	74.04	100.77
31311_at	Homo sapi	78.73	62.91	60.84	72.98	72.9	79.39	85.52	82.57	69.69	63.72	64.29	62.85	67.58
31312_at	potassium	66.65	59.46	55.47	61.75	69.92	75.28	85.53	97.91	69.92	74.77	71.83	58.17	72.15
31313_at	mannosyl	115.33	95.51	84.48	94.99	109.04	105.05	118.68	106.76	142.88	103.72	106.19	98.58	104.13
31314_at	bone morph	71.89	36.24	41.86	46.99	45.94	46.67	67.56	66.14	53.95	40.97	47.96	43.63	53.54
31315_at	immunoglob	103.99	88.27	83.81	81.81	254.63	87.12	99.11	109.56	86.37	75.03	74.97	69.02	97.22
31316_at	Human vac	16.79	10.08	9.53	16.48	11.98	12.8	16.7	18.76	11.25	12.09	18.89	10.81	19.49
31317_r_a	Human unj	316.75	269.61	254.92	352.61	342.4	327.12	366.39	346	308.43	279.81	312.4	318.06	334.27
31318_at	Stem cell f	32.68	19.79	27.45	29.56	28.34	26.55	38.04	41.05	31.91	22.76	23.58	28.29	22.61
31319_at	Cluster Inc	252.78	441.07	143.32	400.01	373.4	105.06	105.72	87.02	110.75	161.69	84.88	240.91	210.54
31320_at	Cluster Inc	101.42	89.07	79.51	100.69	120.06	116.74	121.41	134.74	131.36	137.4	114.15	119.89	126.74
31321_at	Cluster Inc	112.27	62.17	62.44	80.17	110.97	53.89	55.04	55.16	63.37	54.35	57.79	48.07	47.89
31322_at	Cluster Inc	44.15	52.5	44.8	46.25	55.96	50.01	53.2	52.24	62.16	49.94	47.24	40.64	50.1
31323_r_a	Glutamate	141.44	177.7	138.58	142.61	167.28	169.49	199.64	185.22	218.79	196.56	150.14	185.24	226.37
31324_at	Cluster Inc	70.87	57.8	61.61	65.93	84.05	106.41	106.73	87.01	112.12	78.47	111	89.08	100.53
31325_at	Cluster Inc	68.63	167.66	69.04	112.84	120.46	126.72	107.04	100	116.83	207.5	125.65	155.19	102.55
31326_at	Cluster Inc	157.67	127.49	123.37	146.18	150.95	159.46	184.08	206.02	182.95	139.01	154.57	143.09	175.27
31327_at	Cluster Inc	35.57	28.17	33.64	32.36	38.76	43.13	40.16	46.8	34.47	33.71	25.74	29.45	37.37
31328_at	solute carr	61.61	48.23	50.76	58	57.58	57.91	69.91	72.34	70.29	54.98	59.74	45.55	63.02
31329_at	Human pur	12.23	18.91	15.36	19.99	21.15	15.9	20.76	22.26	16.15	28.86	13.59	16.06	23.85
31330_at	ribosomal	108.87	133.3	89.84	113.02	147.61	169.87	156.81	136.47	153.07	220.54	220.96	332.11	183
31331_at	surfactant	28.21	17.99	23.56	26.37	30.35	28.84	31.54	35.06	22.53	24.45	23	21.37	30.86
31332_at	RIG-like 1	20.77	18.58	19.03	18.29	20.86	23.56	25.11	24.43	19.3	28.72	17.27	23.18	25.35
31333_at	tolloid-like	22.97	52.9	26.95	41.22	48.38	48.85	42.09	40.13	40.73	89.86	46.96	59.74	40.87
31334_at	G protein-c	98.57	100.16	78.76	119.09	118.58	97.42	110.67	104.95	143.47	111.28	102.88	115.9	133.17
31335_at	clone 1900	65.79	54.3	54.79	57.23	60.75	66.98	72.89	86.97	76.34	57.65	59.83	49.21	70.83
31336_at	Cluster Inc	40.97	26.15	32.55	26.26	33.73	36.15	36.03	34.93	24.12	26.26	22.55	23.64	28.11

从差异表达到生物发现



手动挑选基因来解读，任意性大，需要生物背景知识



A microarray data example

- Lee et al (2005) compared adipose tissue between **obese and lean** Pima Indians
- Samples were hybridized on HG_U95 Affymetrix microarrays
 - 12639 genes/probe sets
 - **GSE2508** on the GEO database
- Selected the male samples only
 - 10 obese vs. 9 lean



生信思路：搜索和使用公共组学数据

GEO数据库网页: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE2508>

A microarray data example

Microarray profiling of isolated abdominal subcutaneous adipocytes from obese vs non-obese Pima Indians: increased expression of inflammation-related genes

皮下脂肪细胞
中，炎症相关
基因表达升高

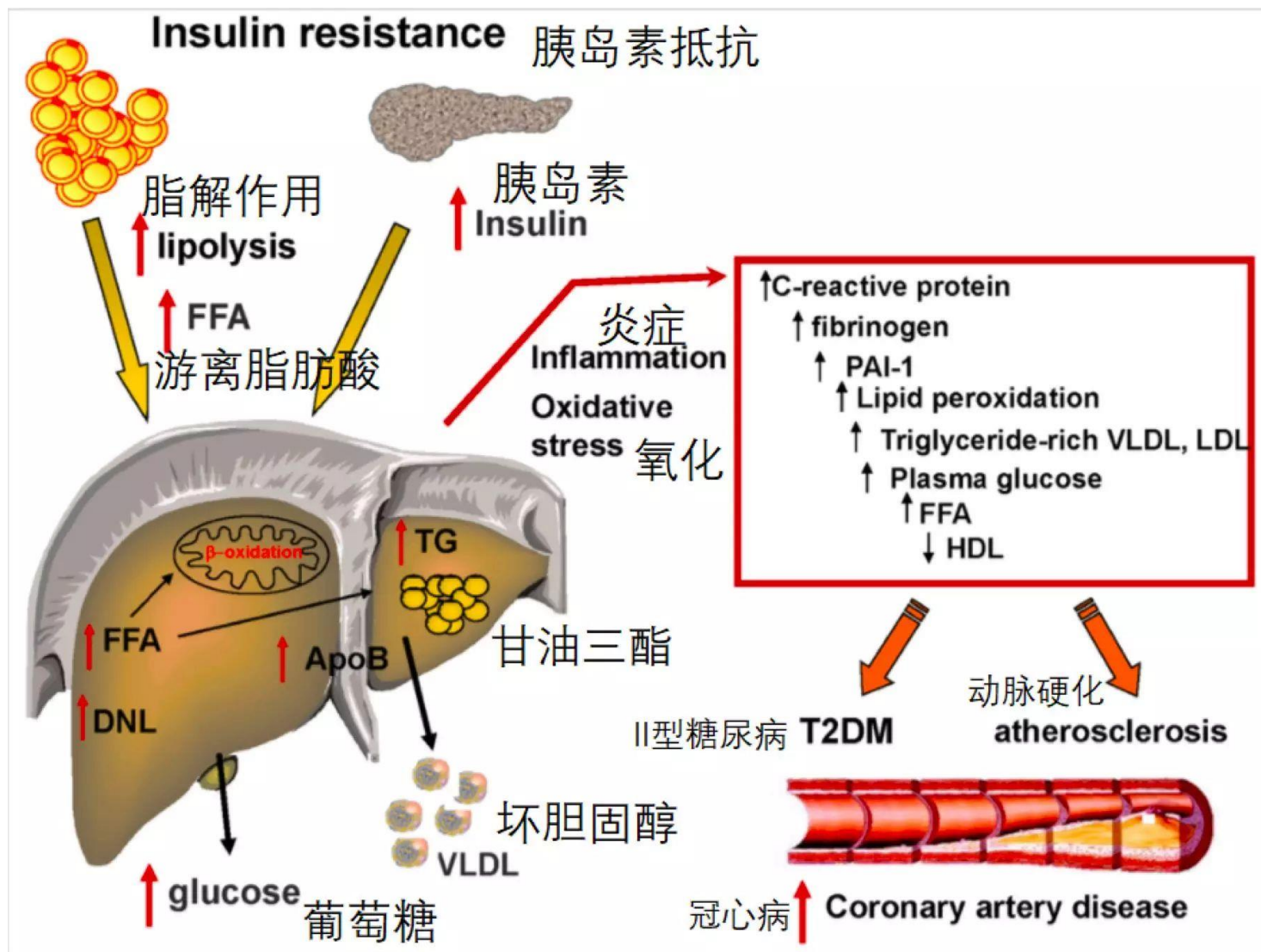
Received: 10 December 2004 / Accepted: 28 April 2005 / Published online: 30 July 2005
© Springer-Verlag 2005

Abstract *Aims/hypothesis:* Obesity increases the risk of developing major diseases such as diabetes and cardiovascular disease. Adipose tissue, particularly adipocytes, may play a major role in the development of obesity and its comorbidities. The aim of this study was to characterise, in adipocytes from obese people, the most differentially expressed genes that might be relevant to the development of obesity. *Methods:* We carried out microarray gene profiling of isolated abdominal subcutaneous adipocytes from 20 non-obese (BMI 25 ± 3 kg/m²) and 19 obese (BMI 55 ± 8 kg/m²) non-diabetic Pima Indians using Affymetrix HG-U95 GeneChip arrays. After data analyses, we measured the transcript levels of selected genes based on their biological functions and chromosomal positions using quantitative real-time PCR. *Results:* The most differentially ex-

pressed genes in adipocytes of obese individuals consisted of 433 upregulated and 244 downregulated genes. Of these, 410 genes could be classified into 20 functional Gene Ontology categories. The analyses indicated that the inflammation/immune response category was over-represented, and that most inflammation-related genes were upregulated in adipocytes of obese subjects. Quantitative real-time PCR confirmed the transcriptional upregulation of representative inflammation-related genes (*CCL2* and *CCL3*) encoding the chemokines monocyte chemoattractant protein-1 and macrophage inflammatory protein 1 α . The differential expression levels of eight positional candidate genes, including inflammation-related *THY1* and *CIQTNF5*, were also confirmed. These genes are located on chromosome 11q22–q24, a region with linkage to obesity in the Pima Indians. *Conclusions/interpretation:* This study provides evidence supporting the active role of mature adipocytes in obesity-related inflammation. It also provides potential candidate genes for susceptibility to obesity.

Electronic supplementary material Supplementary material is available in the online version of this article at <http://dx.doi.org/10.1007/s00125-005-1867-3>.

不抽烟、不喝酒、不爱吃肉，怎么就脂肪肝了？



Differentially expressed genes

Probe Set (Gene)	log2(obese/lean)	pvalue	adj.p
73554_at	1.4971	0.0000	0.0004
91279_at	0.8667	0.0000	0.0017
74099_at	1.0787	0.0000	0.0104
83118_at	-1.2142	0.0000	0.0139
81647_at	1.0362	0.0000	0.0139
84412_at	1.3124	0.0000	0.0222
90585_at	1.9859	0.0000	0.0258
84618_at	-1.6713	0.0000	0.0258
91790_at	1.7293	0.0000	0.0350
80755_at	1.5238	0.0000	0.0351
85539_at	0.9303	0.0000	0.0351
90749_at	1.7093	0.0000	0.0351
74038_at	-1.6451	0.0000	0.0351
79299_at	1.7156	0.0000	0.0351
72962_at	2.1059	0.0000	0.0351
88719_at	-3.1829	0.0000	0.0351
72943_at	-2.0520	0.0000	0.0351
91797_at	1.4676	0.0000	0.0351
78356_at	2.1140	0.0001	0.0359
90268_at	1.6552	0.0001	0.0421

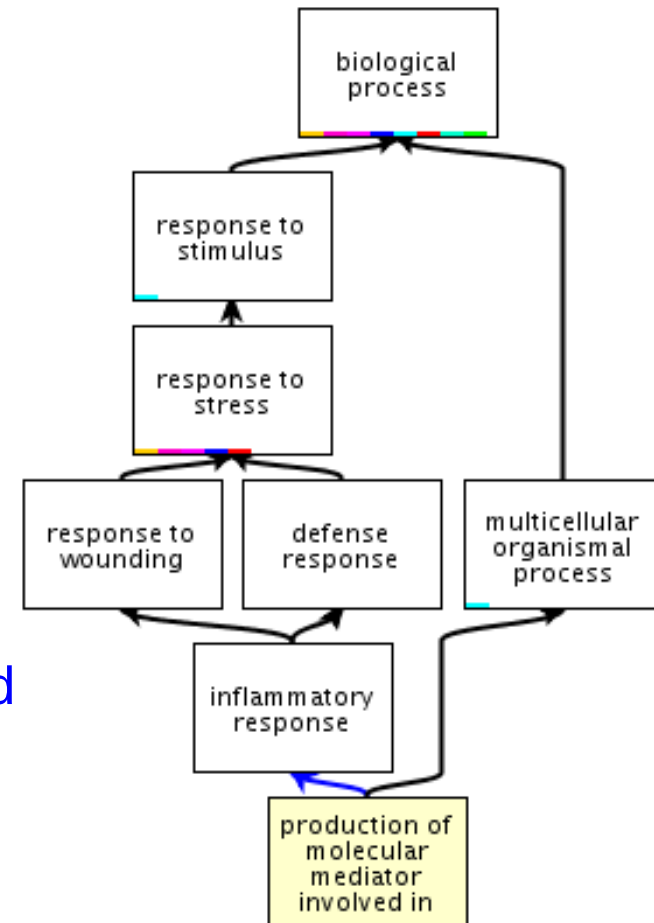
多重检验
校正

Slide source: Claus-D. Mayer

Gene Ontology (GO), 基因功能注释

- The Gene Ontology (GO) database

- <http://www.geneontology.org/>
- GO offers a relational/hierarchical database
- Parent nodes: more general terms
- Child nodes: more specific terms
- At the end of the hierarchy there are genes/proteins
- At the top there are 3 parent nodes: biological process, molecular function and cellular component



- Example

- search the database for the GO term “inflammation”;
- check parent and children nodes.

GO term

inflammatory response

Term Information ?

Accession GO:0006954

Data health ♥

Name inflammatory response

Ontology biological_process

Synonyms Inflammation

Alternate IDs None

Definition The immediate defensive reaction (by vertebrate tissue) to infection or injury caused by chemical or physical agents. The process is characterized by local vasodilation, extravasation of plasma into intercellular spaces and accumulation of white blood cells and macrophages. *Source:* [ISBN:0198506732](#), [GO_REF:0000022](#), [GOC:mtg_15nov05](#)

Comment This term was improved by [GO_REF:0000022](#). It was moved.

History See term [history for GO:0006954](#) at QuickGO

Subset None

Related [Link](#) to all **genes and gene products** annotated to inflammatory response.
[Link](#) to all direct and indirect **annotations** to inflammatory response.
[Link](#) to all direct and indirect **annotations download** (limited to first 10,000) for inflammatory response.

<http://amigo.geneontology.org/amigo/term/GO:0006954>

点击下方Graph views标签 -> QuickGO

GO term structure

Term Lineage

Switch to viewing term parents, siblings and children

Filter tree view ?

Filter Gene Product Counts

Data source

Species

All
AspGD
CGD
dictyBase

All
Anaplasma phagocy...
Arabidopsis thaliana
Bacillus anthraci...

View Options

Tree view

all : all [377382 gene products]

+ 1 GO:0008150 : biological_process [270820 gene products]

+ 1 GO:0050896 : response to stimulus [30457 gene products]

+ 1 GO:0009605 : response to external stimulus [16147 gene products]

+ 1 GO:0009611 : response to wounding [2289 gene products]

+ 1 GO:0006954 : inflammatory response [1173 gene products]

+ 1 GO:0002526 : acute inflammatory response [427 gene products]

+ 1 GO:0002532 : production of molecular mediator of acute inflammatory response [44 gene products]

+ 1 GO:0006950 : response to stress [16147 gene products]

+ 1 GO:0006952 : defense response [4501 gene products]

+ 1 GO:0006954 : inflammatory response [1173 gene products]

+ 1 GO:0002526 : acute inflammatory response [427 gene products]

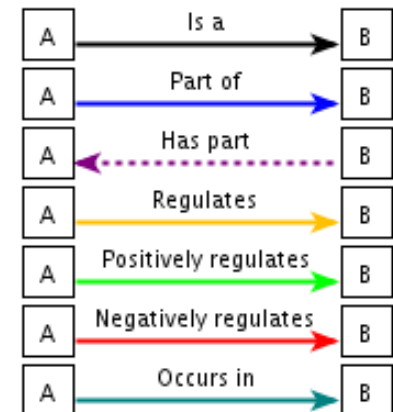
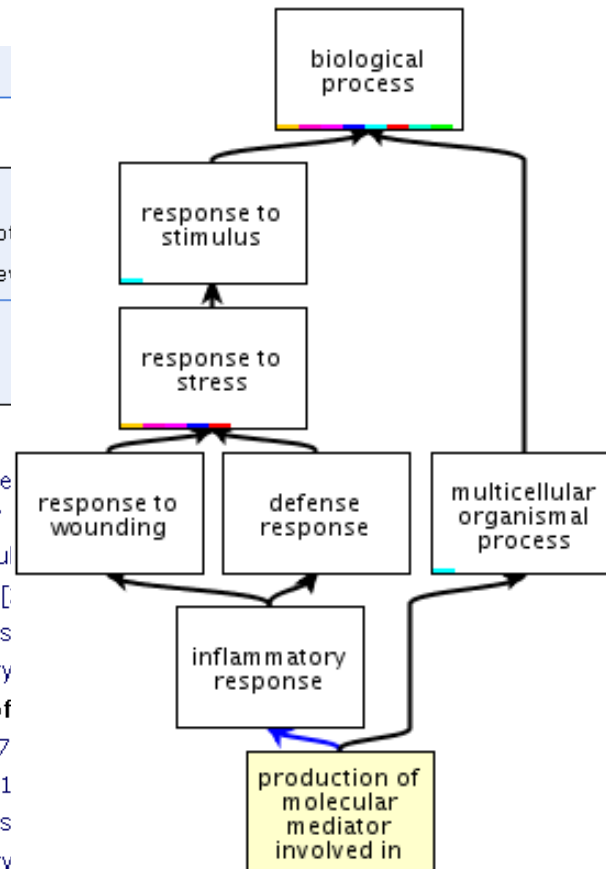
+ 1 GO:0002532 : production of molecular mediator of acute inflammatory response [44 gene products]

+ 1 GO:0009611 : response to wounding [2289 gene products]

+ 1 GO:0006954 : inflammatory response [1173 gene products]

+ 1 GO:0002526 : acute inflammatory response [427 gene products]

+ 1 GO:0002532 : production of molecular mediator of acute inflammatory response [44 gene products]



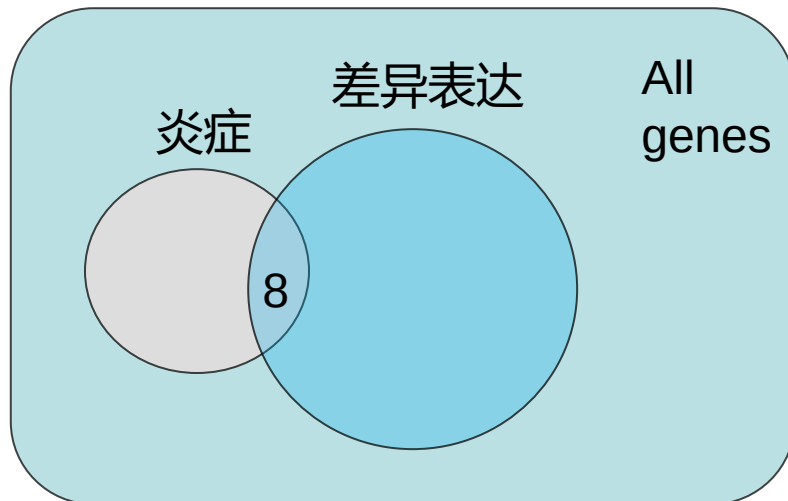
Genes annotated by GO terms

Probe Set ID	Gene Symbol	Gene Title	go biological process term	go molecular function term	log.ratio	pvalue	adj.p
73554_at	CCDC80	coiled-coil domain contain	---	---	1.4971	0.0000	0.0004
91279_at	C1QTNF5	C1q and tumor necrosis f	visual perception	embryonic development	0.8667	0.0000	0.0017
74099_at	---	---	---	---	1.0787	0.0000	0.0104
83118_at	RNF125	ring finger protein 125	immune response	modulation of protein binding	-1.2142	0.0000	0.0139
81647_at	---	---	---	---	1.0362	0.0000	0.0139
84412_at	SYNPO2	synaptopodin 2	---	actin binding	1.3124	0.0000	0.0222
90585_at	C15orf59	chromosome 15 open reading frame	---	---	1.9859	0.0000	0.0258
84618_at	C12orf39	chromosome 12 open reading frame	---	---	-1.6713	0.0000	0.0258
91790_at	MYEOV	myeloma overexpressed	---	---	1.7293	0.0000	0.0350
80755_at	MYOF	myoferlin	muscle contraction	protein binding	1.5238	0.0000	0.0351
85539_at	PLEKHH1	pleckstrin homology domain	---	binding	0.9303	0.0000	0.0351
90749_at	SERPINB9	serpin peptidase inhibitor, anti-apoptosis	signal transduction	endopeptidase inhibitor activity	1.7093	0.0000	0.0351
74038_at	---	---	---	---	-1.6451	0.0000	0.0351
79299_at	---	---	---	---	1.7156	0.0000	0.0351
72962_at	BCAT1	branched chain aminotransferase	G1/S transition of mitotic cell cycle	catalytic activity	2.1059	0.0000	0.0351
88719_at	C12orf39	chromosome 12 open reading frame	---	---	-3.1829	0.0000	0.0351
72943_at	---	---	---	---	-2.0520	0.0000	0.0351
91797_at	LRRC16A	leucine rich repeat containing	---	---	1.4676	0.0000	0.0351
78356_at	TRDN	triadin	muscle contraction	receptor binding	2.1140	0.0001	0.0359
90268_at	C5orf23	chromosome 5 open reading frame	---	---	1.6552	0.0001	0.0421

生信思路：GO term可以帮助理解基因功能，但如何利用它们做大规模计算分析？

Gene function enrichment analysis

- A GO group of 96 genes (感兴趣的基因通路, 如炎症)
- 1272 out of 12639 genes are differentially expressed (差异表达的基因) in this data set (10.1%)
- 问题: 差异表达基因是否与炎症通路显著相关 (8个共同基因) ?
- If we randomly pick 96 genes, how many are differentially expressed?
 - we would expect $96 \times 1272 / 12639 = 9.7$ genes “by chance” (we observed 8)
- A p-value can be calculated based on a 2x2 contingency table (列联表)



	炎症	非炎症	
差异表达	8	1264	1272
非差异表达	88	11 279	11367
	96	12543	12639

每个基因按两种方式分类

R functions for count data

- Association between the counts of two **categorical variables** (分类变量) .
 - Regression: association between two **numerical variables**
- R functions to analyze tabular count data
 - `prop.test()` 比例比较
 - `binom.test()` 二项分布
 - `chisq.test()` 卡方检验
 - `fisher.test()` Fisher检验

Tests for association with R

- Compare proportions
- Chi-square test
- Fisher's exact test

	In GS	Not in GS
sig	8	1264
non-sig	88	11 279

```
> x <- matrix(c(8, 1264, 88, 11279), nrow=2, byrow=T)
> x
      [,1] [,2]
[1,]    8 1264
[2,]   88 11279
> chisq.test(x)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data:  x
X-squared = 0.15645, df = 1, p-value = 0.6924
```

可以很方便地在R中计算，但要充分理解背后的统计方法，才能正确使用和解释结果

对多个GO term做功能富集分析

Categories	Upregulated in obese subjects	Downregulated in obese subjects	<i>p</i> value of category over-representation ^a	<i>p</i> value of difference in the number of upregulated vs down regulated genes ^a
Inflammation/ immune response	52	2	0.0008	<0.00001
Signalling	25	17	0.001	
Transcription regulation	12	22	0.002	
Cell cycle control/cell proliferation	24	8		0.002
Cell adhesion	26	2		<0.00001
Transport/carrier	18	14		
Structural protein/ cytoskeleton organisation	21	2		<0.00001
Apoptosis	14	6		0.04
Protein biosynthesis/ folding	6	11		

Gene Ontology Biological Process classification of the 410 most differentially expressed genes as indicated by the microarray data

二项分布举例

这是一个二项分布计算网站。如果每人放弃offer概率是0.2 (15/76) , 86人 (n) 中10人 (x) 或更多人放弃的概率是?

Binomial Distribution Calculator

Use this binomial probability calculator to easily calculate binomial cumulative distribution function and probability mass given the probability on a single trial, the number of trials and events. The calculator can also solve for the number of trials required.

Calculate  ☒ Probability ☐ Trials required

Probability (p) 

Number of trials (n) 

Number of events (x) 

 Calculate

✓ Calculation results		
Probability of exactly 10 events	0.01564054809109 (1.56%)	
Probability of $X \leq 10$ events	0.02977062542830 (2.98%)	
Probability of $X > 10$ events	0.97022937457170 (97.02%)	
Probability of $X < 10$ events	0.01413007733721 (1.41%)	
Probability of $X \geq 10$ events	0.98586992266279 (98.59%)	

Share calculator:



Share your result!



get direct link 

Embed this tool:

get code 

<https://www.gigacalculator.com/calculators/binomial-probability-calculator.php>

本节内容

- 分类数据举例
- 基因功能富集分析
- ➡ • 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

用列联表检验分类变量的相关性

		Outcome		
		Disease	Non-Disease	
Risk Factor	Yes	50	0	Perfect Positive Association between the risk factor and outcome ←
	No	0	50	

35	15
15	35

↓
Some Degree of Association
(risk factor is deleterious)

25	25
25	25

↓
No association at all

统计方法类型（预测 / 相关性检验）

Y变量 X变量	数值（定量）	类别（定性）
数值	线性回归	t-检验，方差分析
类别	分类方法	列联表

2x2 Contingency Table (列联表)

- The table shows the data from a study of 91 patients who had a myocardial infarction (Snow 1965).
- One variable is **treatment** (propranolol versus a placebo), and the other is **outcome** (survival for at least 28 days versus death within 28 days).

问题：新药是否比安慰剂显著好？
(结果是否由随机因素造成)

	<u>OUTCOME</u>		Total
	Survival for at least 28 days	Death	
<u>Treatment</u>			
Propranolol (心得安)	38	7	45
Placebo	29	17	46
Total	67	24	91

The hypotheses for two-way tables

- The null hypothesis H_0 is simply that ***there is no association*** between the row and column variables.
 - The method of treating the myocardial infarction patients **did not influence** the proportion of patients who survived for at least 28 days.
- The alternative hypothesis H_a is that ***there is an association*** between the two variables.
 - It doesn't specify a particular causal direction
 - Assume the outcome (survival time) **depends** on the treatment.

统计检验思路:

1. 把无效果或不感兴趣的结论作为**零假设**
2. 设计**统计量**, 计算零假设下产生观测数据或更极端数据的概率 (p-value)
3. 如果这个概率很小 (可能零假设、模型或数据错了), 则推翻零假设

Calculation of Expected Cell Count

- The expected cell counts under H_0

$$E_{1,1} = \frac{\text{Row}_1 \text{ Total} \times \text{Column}_1 \text{ Total}}{\text{Study Total}}$$

按总体列比例，
把每一行总数分
配到各列

考虑所有病人，长期生存和短期生存人数的比例约是3: 1

如果零假设成立（用药和预后生存无关），每行的计数都应该约是这个比例

<u>Treatment</u>	<u>OUTCOME</u>		Total
	Survival for at least 28 days	Death	
Propranolol	33.13	11.87	45
Placebo	33.87	12.13	46
Total	67	24	91

计算期望值用的5个Total
值中，3个值独立并基于
4个原始数据点； χ^2 统
计量自由度为4-3

Calculation of Expected Cell Count

Observed cell counts:

<u>Treatment</u>		<u>OUTCOME</u>		Total
		Survival for at least 28 days	Death	
	Propranolol	38	7	45
	Placebo	29	17	46
	Total	67	24	91

Expected cell counts (零假设下的预期结果) :

<u>Treatment</u>		<u>OUTCOME</u>		Total
		Survival for at least 28 days	Death	
	Propranolol	33.13	11.87	45
	Placebo	33.87	12.13	46
	Total	67	24	91

统计量：Chi-Square (χ^2) Test Statistic

Calculation of the chi-square (χ^2) value

$$\begin{aligned}\chi^2 &= \sum \left[\frac{(O - E)^2}{E} \right] \\ &= \frac{(38 - 33.13)^2}{33.13} + \frac{(7 - 11.87)^2}{11.87} + \frac{(29 - 33.87)^2}{33.87} + \frac{(17 - 12.13)^2}{12.13} \\ &= 5.38\end{aligned}$$

$$df = (R - 1)(C - 1) = 1$$

$$p\text{-value} < 0.025$$

Interpretation:

The results noted in this 2×2 table are statistically significant.

That is, it is highly improbable (only 1 chance in about 50 of being wrong) that the difference is observed under H_0 . Therefore the investigator can reject the null hypothesis of independence and accept the alternative hypothesis that propranolol does affect the outcome of myocardial infarction.

结果解读：如果零假设成立， χ^2 统计量近似服从 χ^2 分布，在大概率情况下，统计量将取值小，p-value不显著

The process of comparing the observed counts with the expected counts is called a **goodness-of-fit test** (拟合优度检验)

The Chi-Square (χ^2) Test p-value

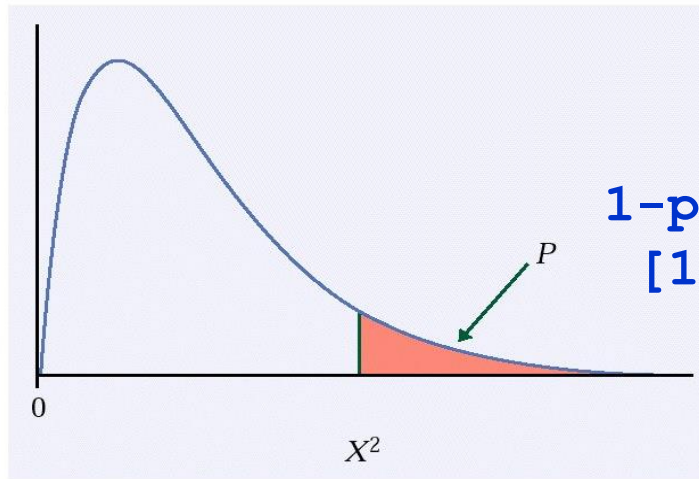
CHI-SQUARE TEST FOR TWO-WAY TABLES

The null hypothesis H_0 is that there is no association between the row and column variables in a two-way table. The alternative is that these variables are related.

If H_0 is true, the chi-square statistic X^2 has approximately a χ^2 distribution with $(r-1)(c-1)$ degrees of freedom.

The P -value for the chi-square test is

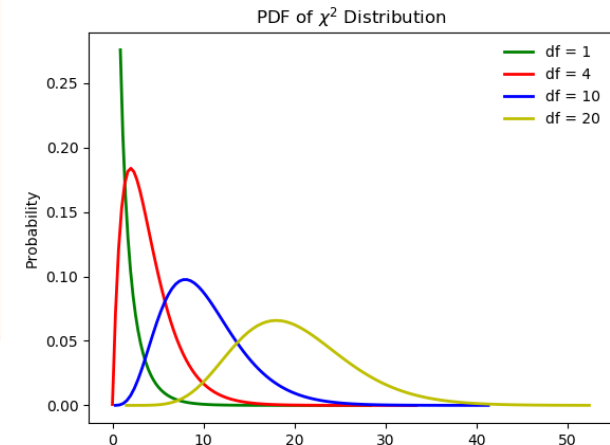
$$P(\chi^2 \geq X^2)$$



```
1-pchisq(5.38,1)  
[1] 0.02036888
```

where χ^2 is a random variable having the $\chi^2(df)$ distribution with $df = (r-1)(c-1)$.

如果 χ^2 统计量大，
则p-value显著，说明
零假设下产生这样
极端的数据的概率
很小，所以推翻
零假设



The Chi-Square (χ^2) Test Statistic

The chi-square statistic is a measure of how much the observed cell counts in a two-way table differ from the expected cell counts under H_0 .

$$\chi^2 = \sum (\text{observed count} - \text{expected count})^2 / \text{expected count}$$

Degrees of freedom (df) = $(r - 1) \times (c - 1)$

- r is the number of rows, c is the number of columns, and the sum (Σ) is over all the $r \times c$ cells in the table.
- For 2 x 2 tables, need all the expected cell counts ≥ 5 .
- For tables larger than 2 x 2, need the average of the expected cell counts is ≥ 5 and the smallest expected cell count is ≥ 1
- If the expected counts and observed counts are very different, a large chi-square value will result, providing evidence AGAINST the null hypothesis.

What do we do if the expected values in any of the cells in a 2x2 table is below 5?

	men	women	total
dieting	1	9	10
not dieting	11	3	14
totals	12	12	24

每个格子数值小时， χ^2 统计量不近似服从 χ^2 分布，会导致错误P值

We guess that the proportion of dieting individuals is higher among women than among men, and we want to test whether any difference of proportions that we observe is significant.

课后阅读内容：Fisher精确检验

Fisher's exact test (Fisher精确检验)

	men	women	total
dieting	1	9	10
not dieting	11	3	14
totals	12	12	24

The question we ask about these data is: knowing that 10 of these 24 teenagers are dieters, what is the probability that these 10 dieters would be so unevenly distributed between the girls and the boys?

- If we were to choose 10 of the teenagers at random (no relation to dieting), what is the probability that 9 of them would be among the 12 girls, and only 1 from among the 12 boys?

--Fisher's exact test uses hypergeometric distribution (超几何分布) to calculate the “exact” probability of obtaining such set of the values.

Fisher's exact test

	men	women	total
dieting	a	c	$a + c$
not dieting	b	d	$b + d$
totals	$a + b$	$c + d$	n

R.A. Fisher showed that the probability of obtaining any such set of values was given by the **hypergeometric distribution**:

$$p = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}}$$
$$= \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{n!a!b!c!d!}$$

Our example

As extreme
as observed

	men	women	total
dieting	1	9	10
not dieting	11	3	14
totals	12	12	24

$$p = \frac{10!14!12!12!}{24!1!9!11!3!} = 0.00134$$

```
> dhyper(1,12,12,10)
[1] 0.001346076
```

dhyper(x, m, n, k)

x: 取到的白球

m: 白球总数 (男)

n: 黑球总数 (女)

k: 取球总数 (节食)

More
extreme than
observed

	men	women	total
dieting	0	10	10
not dieting	12	2	14
totals	12	12	24

$$p = \frac{10!14!12!12!}{24!0!10!12!2!} = 0.00003$$

```
> dhyper(0,12,12,10)
[1] 3.36519e-05
```

p-value is the probability of observing data as extreme or more extreme if the null hypothesis is true. So the p-value is this problem is 0.00137, rejecting H_0 (no association between gender and dieting).

R fisher.test function

```
> x <- matrix(c(1, 9, 11, 3), nrow=2, byrow=T)
```

```
> x
```

```
      [,1] [,2]  
[1,]     1     9  
[2,]    11     3
```

```
> fisher.test(x)
```

Fisher's Exact Test for Count Data

```
data: x
```

```
p-value = 0.002759
```

```
alternative hypothesis: true odds ratio is not equal to 1
```

```
95 percent confidence interval:
```

```
 0.0006438284 0.4258840381
```

```
sample estimates:
```

```
odds ratio
```

```
0.03723312
```

双侧检验:

```
>dhyper(1,12,12,10)+dhyper(0,12,12,10)  
+dhyper(10,12,12,10)+dhyper(9,12,12,10)  
[1] 0.002759456
```

```
> fisher.test(x, alternative="less")
```

Fisher's Exact Test for Count Data

```
data: x
```

```
p-value = 0.00138
```

```
alternative hypothesis: true odds ratio is less than 1
```

```
95 percent confidence interval:
```

```
 0.00000000 0.3260026
```

```
sample estimates:
```

```
odds ratio
```

```
0.03723312
```

单侧检验:

```
> dhyper(1,12,12,10)+  
dhyper(0,12,12,10)  
[1] 0.001379728
```

More about Fisher's Exact Test

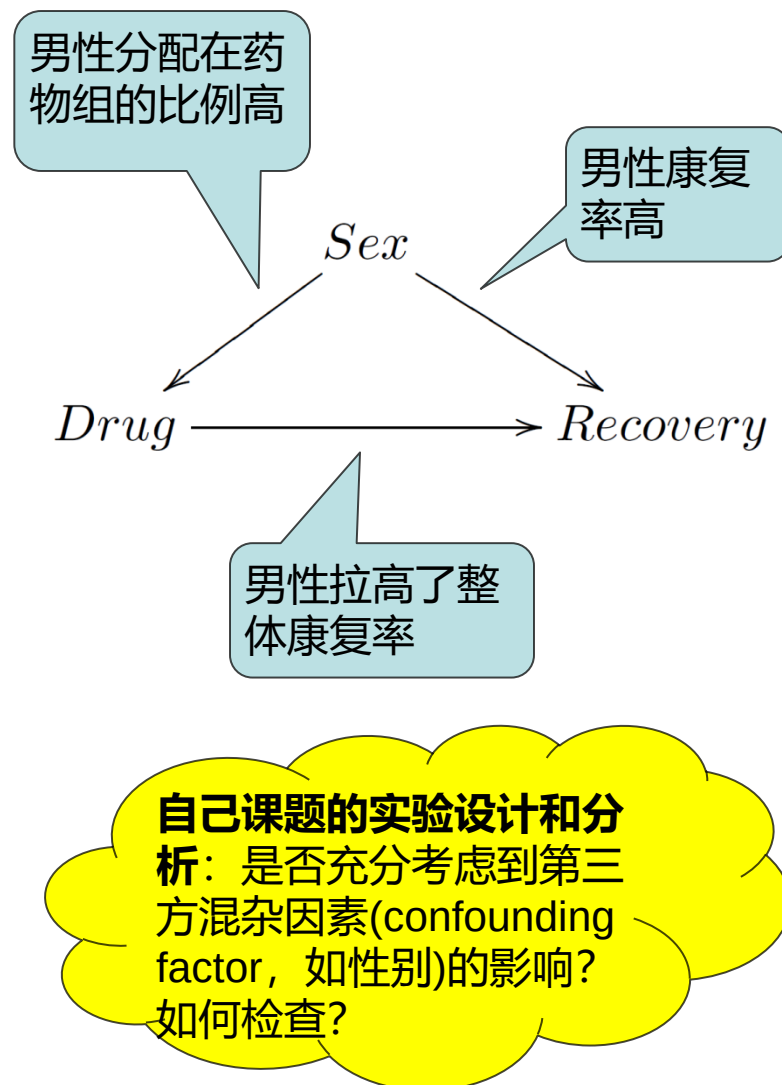
- Used when one or more of the expected counts in a contingency table is small (<2).
 - There's really no lower bound on the amount of data that is needed for Fisher's Exact Test. You can use Fisher's Exact Test when one of the cells in your table has a zero in it.
 - Fisher's Exact Test is also very useful for highly imbalanced tables. If one or two of the cells in a two by two table have numbers in the thousands and one or two of the other cells has numbers less than 5, you can still use Fisher's Exact Test.
- Fisher's Exact Test is based on exact probabilities from a specific distribution (the hypergeometric distribution).
 - Fisher's Exact Test has no formal test statistic and no critical value, and it only gives you a p-value.
- The Chi-square test relies on a large sample approximation. Therefore, you may prefer to use Fishers Exact test in situations where a large sample approximation is inappropriate. (保险做法：两种检验都做一下)

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- 统计**作图**：富集作图
- 富集分析软件

Stat Pitfalls: Yule-Simpson悖论

合并表	康复	未康复	康复率
吃药	20	20	50%
安慰剂	16	24	40%
男性	康复	未康复	康复率
吃药	18	12	60%
安慰剂	7	3	70%
女性	康复	未康复	康复率
吃药	2	8	20%
安慰剂	9	21	30%



研究生录取是否有性别歧视？

UC Berkeley 的统计学家**Peter Bickel** 教授1975 年在Science 上发表文章，报告了Berkeley 研究生院男女录取率的差异：

- 总体上，男性的录取率高于女性
- 然而按照专业分层后，女性的录取率却高于男性
- 女性偏好申请的专业申请人数多、录取率低

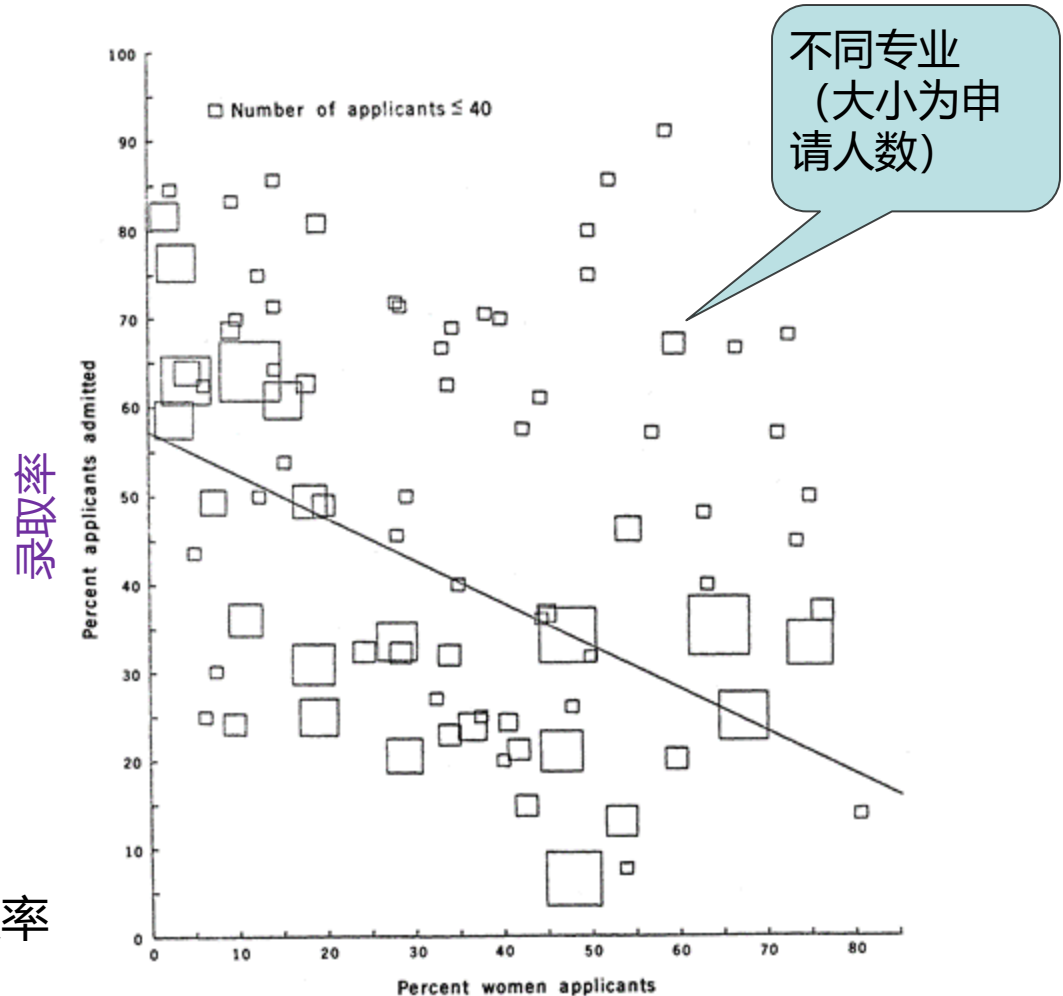
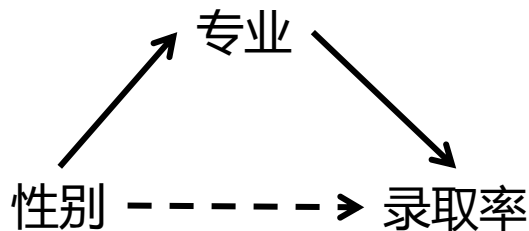


Fig. 1. Proportion of applicants that are women plotted against proportion of applicants admitted, in 85 departments. Size of box indicates relative number of applicants to the department.

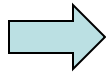
申请者中女性比例

Bickel, P.J. et al, 1975, Sex Bias in Graduate Admissions: Data from Berkeley, 主页: <https://statistics.berkeley.edu/people/peter-bickel>

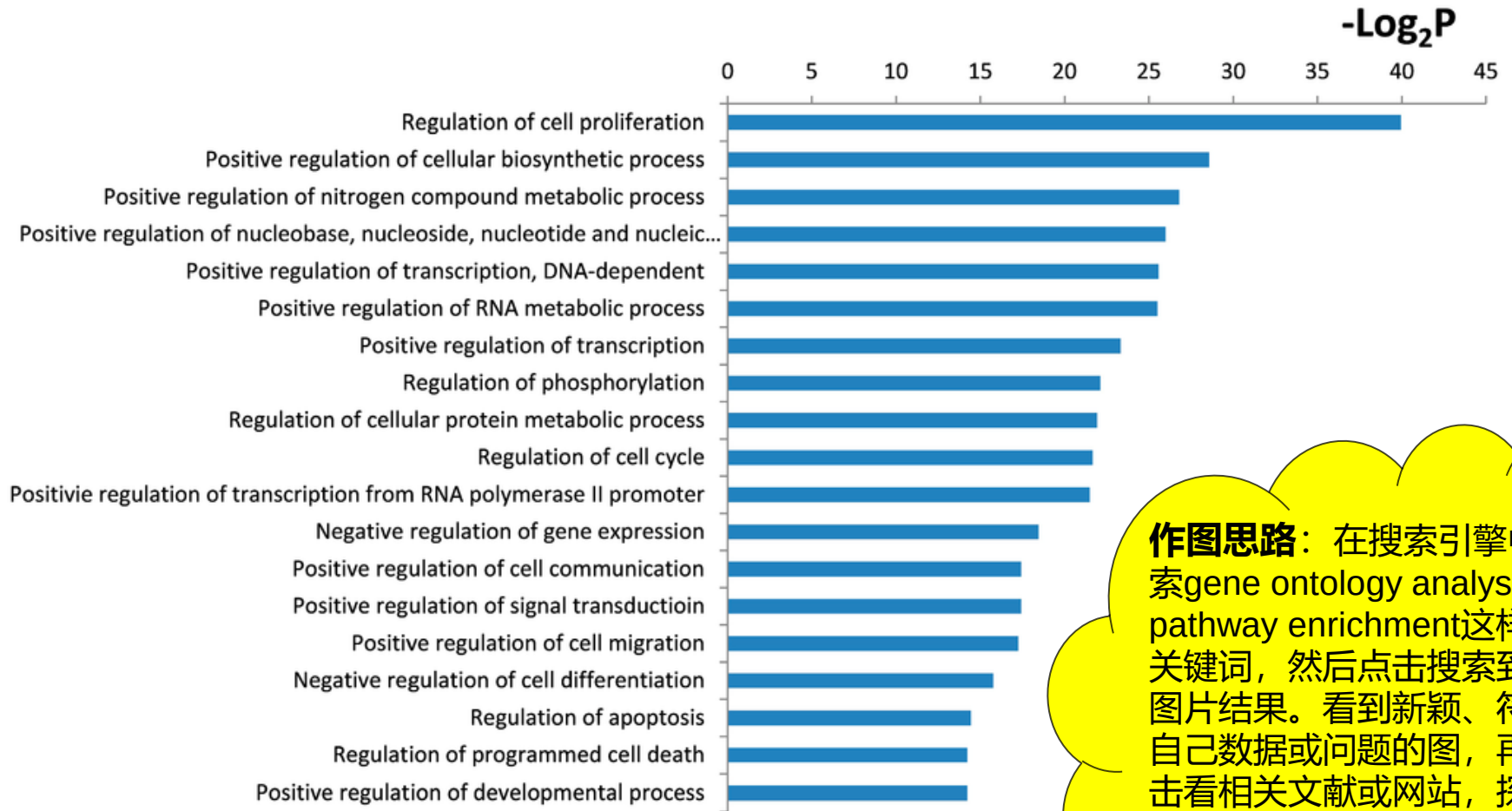


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Stat Graphs: Gene function enrichment



作图思路： 在搜索引擎中搜索gene ontology analysis, pathway enrichment这样的关键词，然后点击搜索到的图片结果。看到新颖、符合自己数据或问题的图，再点击看相关文献或网站，探索这个图是怎么做出来的。

基因集合的功能富集、通路和网络图

<https://mp.weixin.qq.com/s/ZFqfqwmXa5gfl8j7BY49Zg>

P-value plot

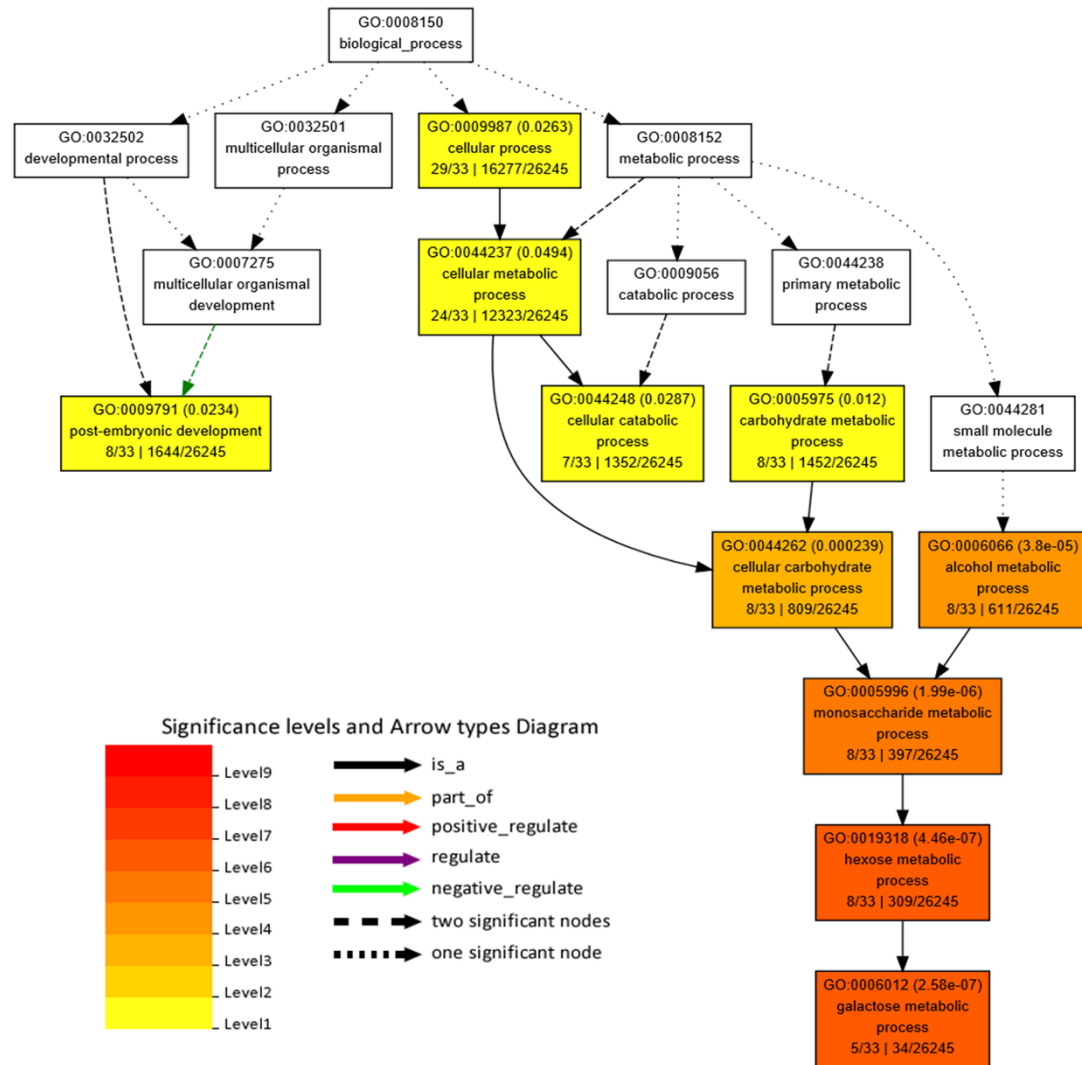
优点:

- 简洁；图比数据表能更直观、快速地传达信息。

缺点:

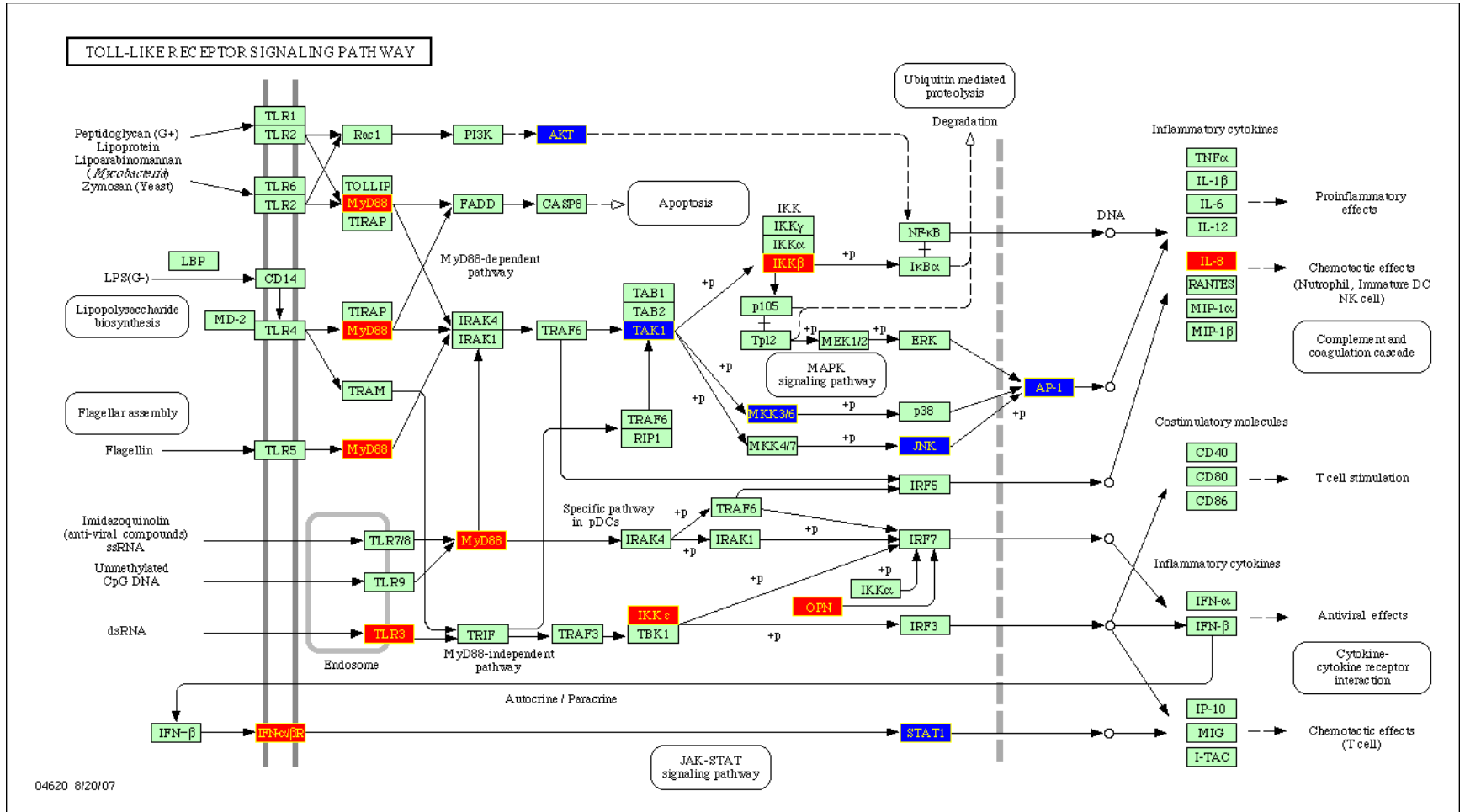
1. 许多富集的GO term含义比较笼统，如transcription, cell cycle，不能帮助更好地聚集研究线索。
 - 这是因为针对2x2列联表的统计检验，如Fisher's test、Chi-square test，如果把列联表的四个数都乘以10，富集比例不变，但p-value会变得更显著（**课后练习**）。这些笼统的GO term集合包含的基因数目多，它们就容易得到显著的p-value。
2. GO terms组成一个类似树状的关系图，terms沿树根到枝叶是从笼统到具体的功能关系，在树中临近的GO terms含义相关，也包含许多共同的基因，所以富集结果中会看到冗余的GO terms富集。
3. 图比较朴素，容易产生审美疲劳。

GO analysis using AgriGO



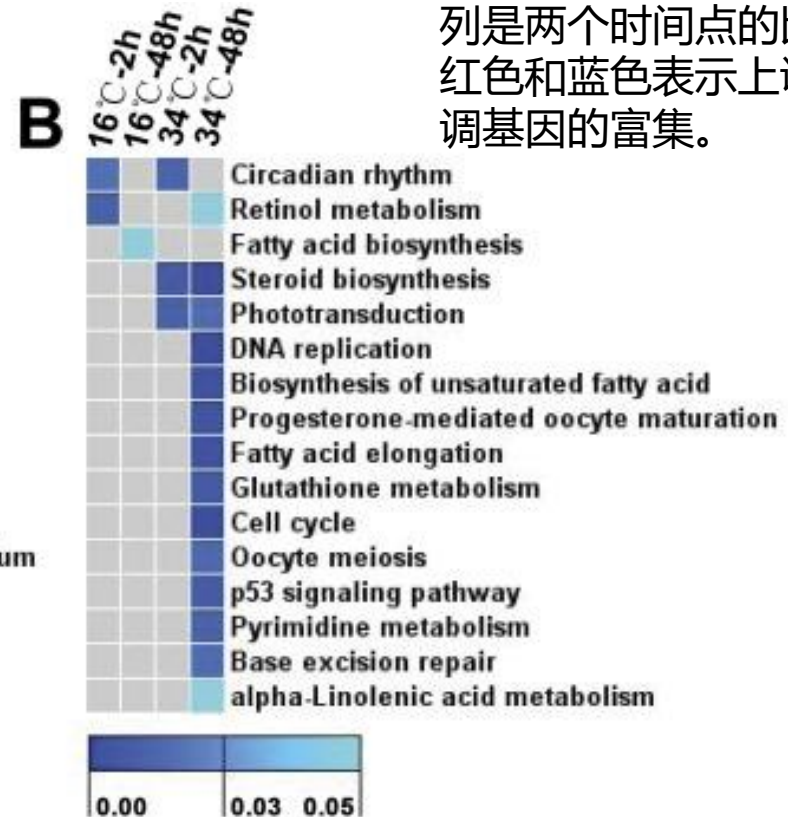
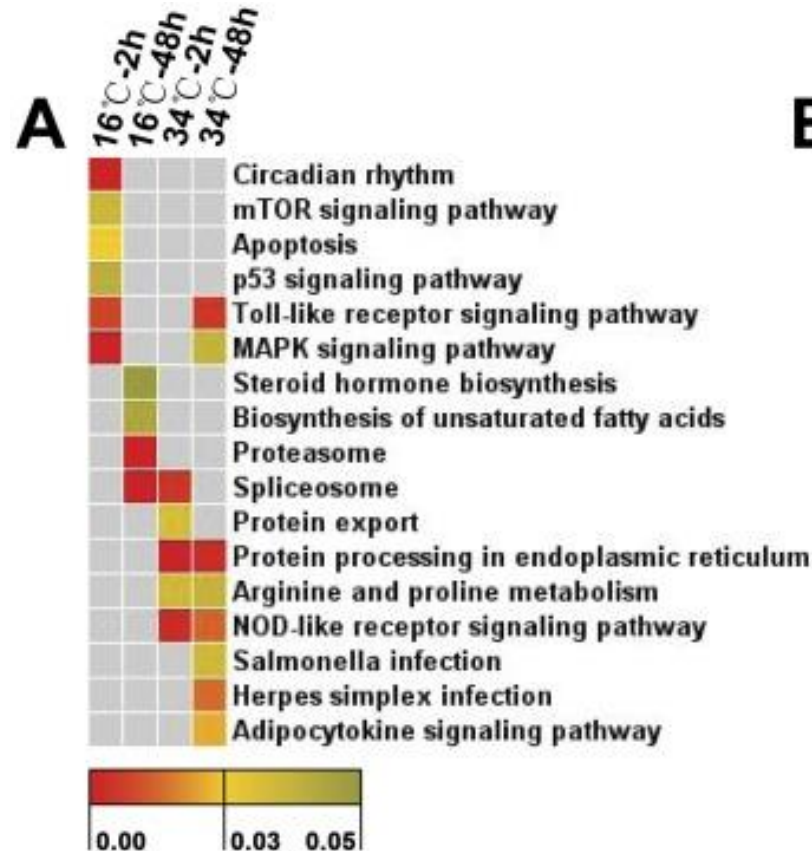
改进1: 把GO term的关系树画出来, 富集的程度用颜色表示, 很直观地看到相邻term富集程度的渐变

KEGG plot



改进2: 在感兴趣的或富集分析得到的生物通路图（如KEGG 通路）中，用颜色标出基因表达变化，帮助看到变化的通路节点或分支

多条件下的富集热图



这里每行是不同通路，每列是两个时间点的比较，红色和蓝色表示上调和下调基因的富集。

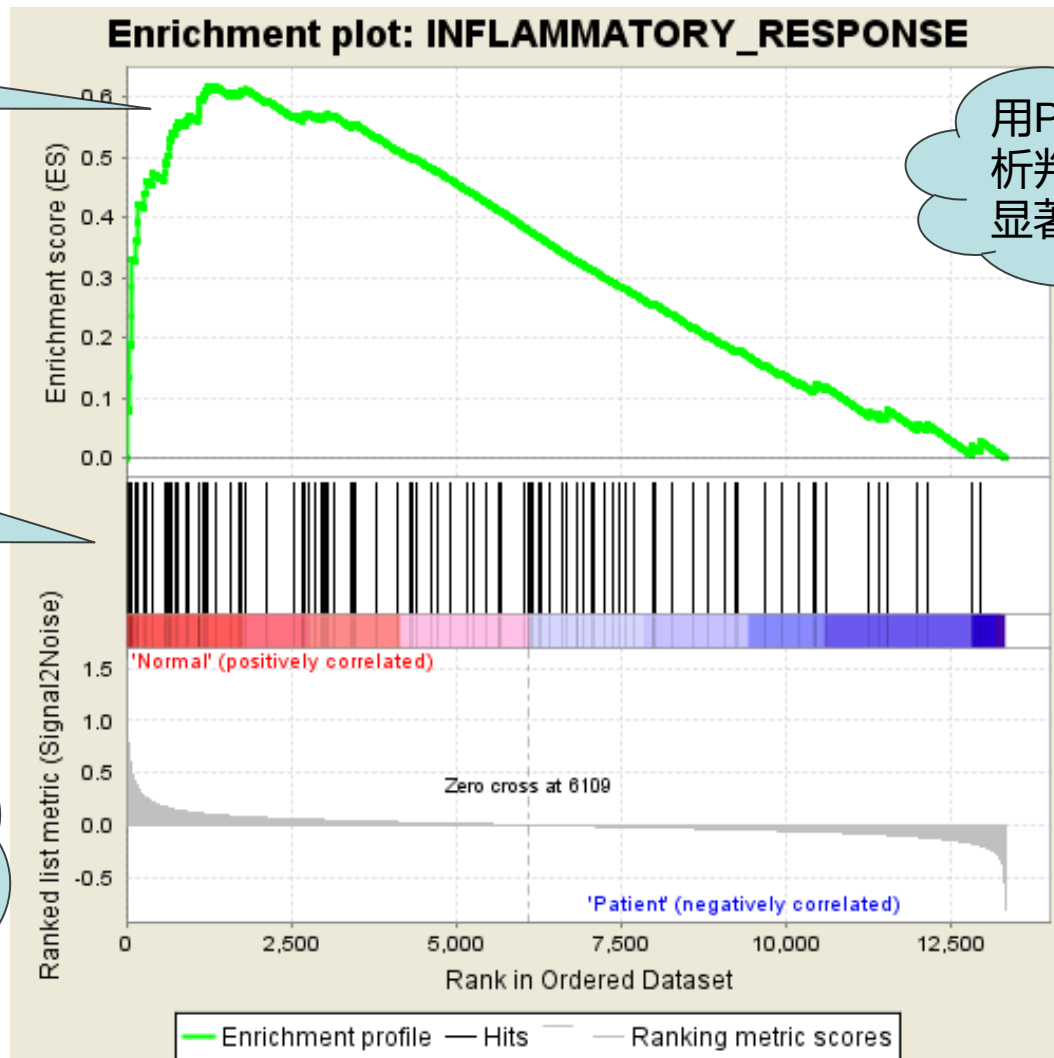
改进3：相比GO terms，生物通路数量少一些，相互也更独立些，所以可以用热图同时展示多个通路在不同条件下差异表达基因的富集情况，看到通路富集随时间的变化

Gene set enrichment analysis (GSEA)

富集打分

感兴趣的基因功能/通路列表，如细胞周期基因

优点：不用特定差异表达阈值，使用所有基因差异表达的信息



用Permutation分析判断富集打分的显著性

所有基因的排序，例如按差异表达

PNAS 2005 Gene set enrichment analysis A knowledge-based approach for interpreting genome-wide expression profiles

软件主页: <http://www.broadinstitute.org/gsea/>

算法介绍 (中文): http://blog.csdn.net/qq_29300341/article/details/52956052

文献举例

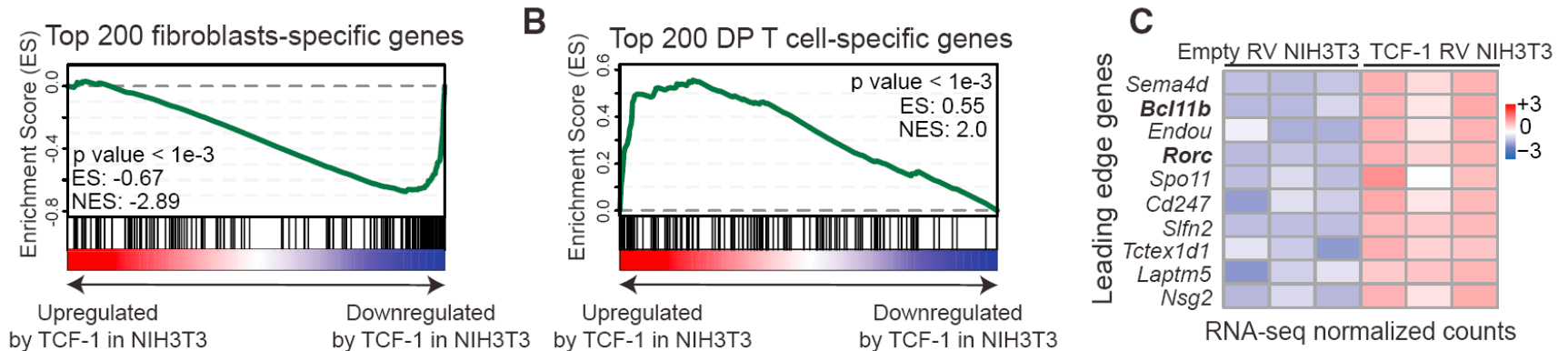
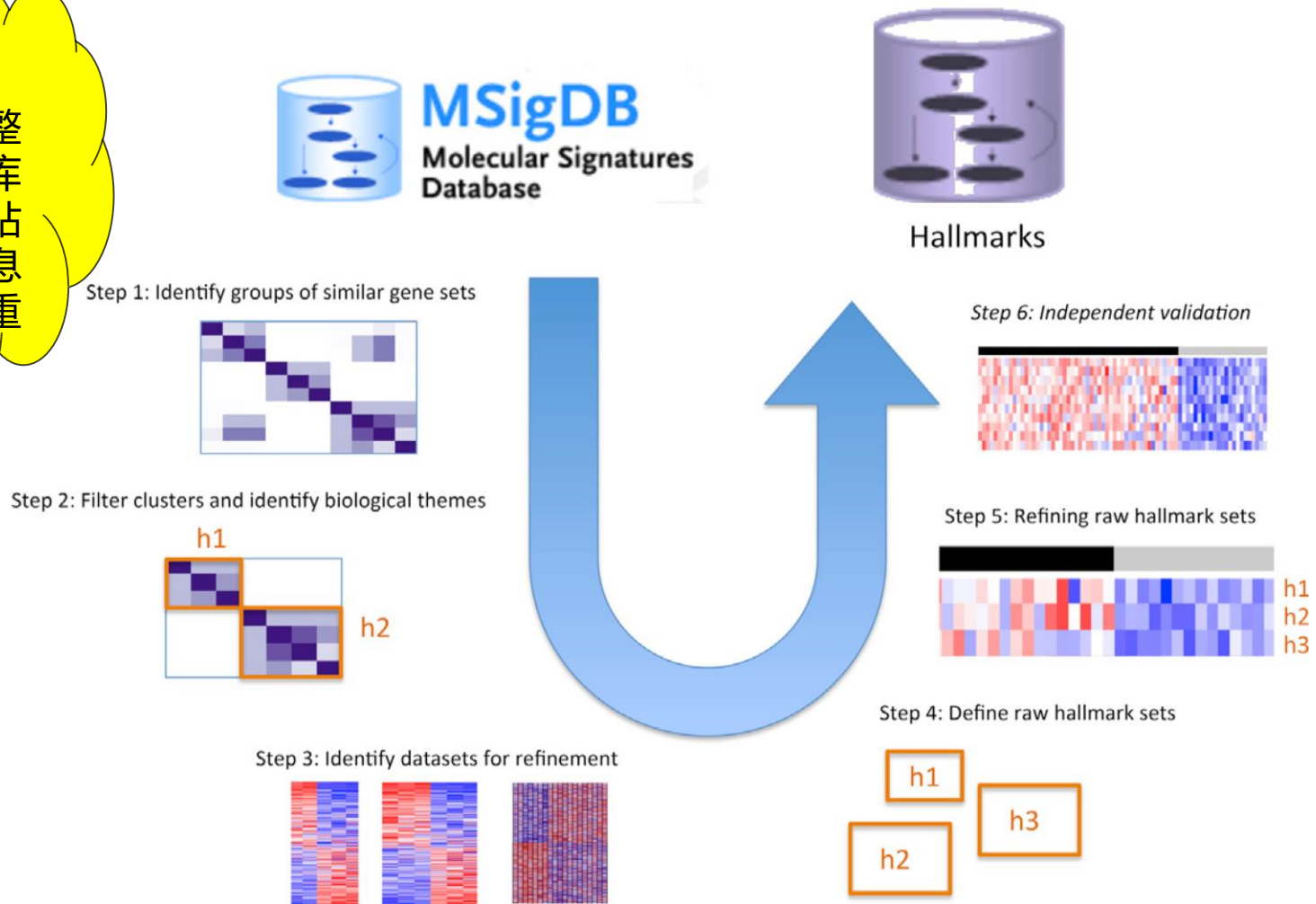


Figure 6. T Cell-Specific Genes Innately Repressed in Fibroblasts Are Upregulated by TCF-1 (see also Figure S6)
 (A–C) Three replicates of RNA-seq in TCF-1 RV and Empty RV NIH 3T3 cells were generated to assess the effects on gene expression. DESeq2 (fold-change > 2 and p adj < 0.05) facilitated the differential gene-expression analysis resulting in 1,295 down- and 1,477 upregulated genes in TCF-1 RV NIH 3T3 cells (see Figures S6A and S6B). Genes located in non-canonical chromosomes were removed from the lists. In addition, we applied differential gene-expression analysis between Empty RV and DP T cells to establish cell specific gene expression (see STAR Methods), which facilitated GSEA analysis of DEGs in TCF-1 RV NIH 3T3 cells on the fibroblast gene set (A) and the T cell gene set (B). Leading edge analysis (C) in top T cell genes.

The Molecular Signatures Database Hallmark Gene Sets

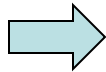
生信思路：
数据收集整理、数据库和分析网站是生物信息学的一个重要分支



Generated 50 “hallmark” gene sets from the Molecular Signature Database (MSigDB).
The hallmarks **reduce redundancy and produce more robust enrichment analysis results.**

本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件



KOBAS web server

KOBAS 2.0

Run KOBAS 2.0

Annotate

Identify

Advanced KOBAS

User Space

Analysis history

Download

Download

Help

Tutorial

Feedback

Contact

Users

My Account

Logout

Annotate - Annotates queries with KEGG GENES or KO terms, also annotates with pathway and disease information

The types of queries can be a set of protein or nucleotide sequences in FASTA format, the tabular format output of blast program, or a list of IDs (can be Entrez Gene ID, UniProtKB AC, or GI). Given queries inputted by users, Annotate assigns KEGG GENES or KO terms based on sequence similarity search, parsing blast output or ID mapping. If the type of input file is sequence, the maximum number of sequences is **500**. If you want to annotate more sequences, you need to download sequences file of the desired species and run BLAST locally. Then do annotation on KOBAS with the tabular output file of BLAST. And you also can download standalone version to run locally.

After clicking the "Example" hyperlink, all the form will be filled automatically, and you can execute the program by simply clicking the 'Run' button.

[FASTA example](#)

[Blasttab example](#)

[Gene id example](#)

Input

Use which file source?

☒ A file on the server

☐ Paste from clipboard

☐ Upload a file from local disk

File type:

Entrez Gene ID

Which file contains the genes you want to annotate?:

|-core.list

To which species do you want to map the genes to KEGG?

Pathway

Disease

Database

URL

KEGG PATHWAY

<http://www.genome.jp/kegg/pathway.html>

PID

<http://pid.nci.nih.gov/>

BioCarta (from PID)

<http://www.biocarta.com/>

Reactome

<http://www.reactome.org/>

BioCyc

<http://biocyc.org/>

PANTHER

<http://www.pantherdb.org/>

Gene Ontology

<http://www.geneontology.org/>

OMIM

<http://www.ncbi.nlm.nih.gov/omim/>

KEGG DISEASE

<http://www.genome.jp/kegg/disease/>

FunDO

<http://django.nubic.northwestern.edu/fundo/>

GAD

<http://geneticassociationdb.nih.gov/>

NHGRI

GWAS

<http://www.genome.gov/gwastudies/>

Catalog

Output

Save result in directory:

Output file:

北京大学生科院魏丽萍组, <http://kobas.cbi.pku.edu.cn/>

用网站做分析：基因富集分析



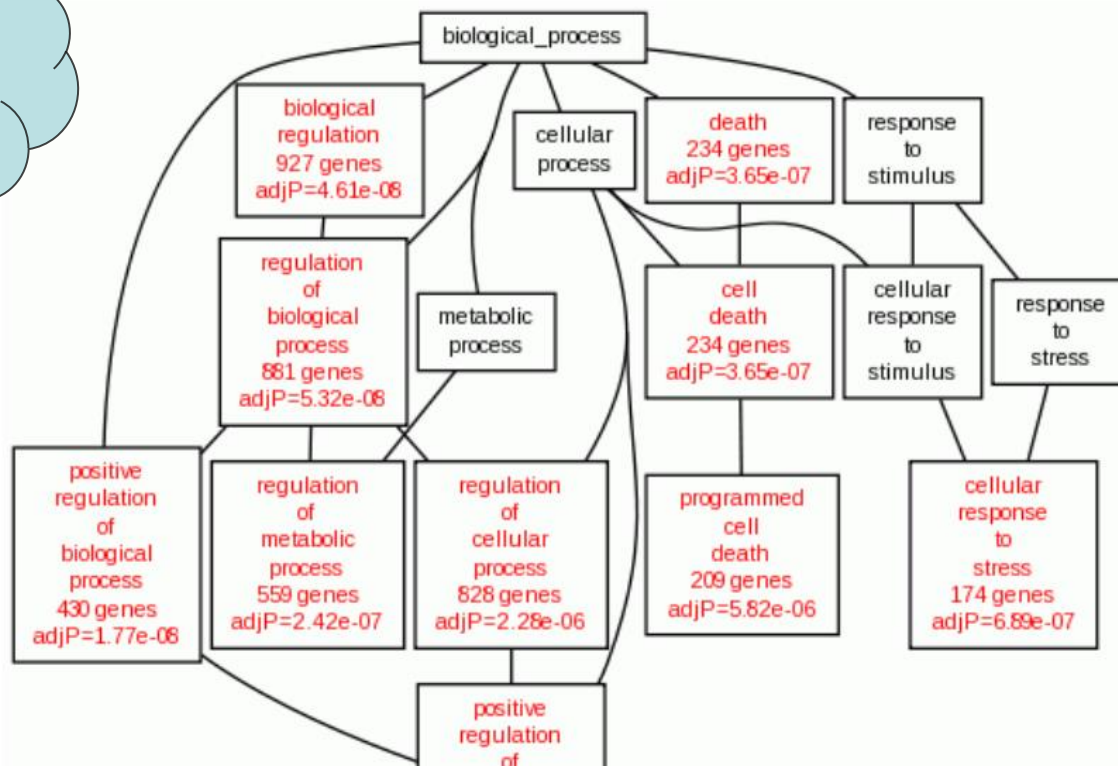
WEB-based GENE SeT AnaLysis Toolkit

Translating gene lists into biological insights...

WebGestalt

CACNA1G
LRRC75A-
AS1
ATXN3
ST3GAL3
DISC1
PGAP2
PLAGL1
MUC1
PDE9A
UBE2V1
VEGFA
BCL2L11
ZMYND8
CTCFL
CHN2
IKZF3
ZNF133
ZNF302

问题：MYC
蛋白可能调控
哪些基因
通路？



MYC 结合位点基因
(MYC binding.xlsx)

WebGestalt 演示

← → ↻ 🏠 ⓘ www.webgestalt.org/option.php

» Basic Parameters

Select Organism of Interest ⓘ

hsapiens

Select Method of Interest ⓘ

Overrepresentation Enrichment Analysis (ORA)

Select Functional Database ⓘ

geneontology

Biological_Process

Gene List

Select Gene ID Type ⓘ

genesymbol

选择文件 未选择任何文件 Reset

Upload Gene List (max size: 5 MB) ⓘ

OR

CACNA1G
LRRC75A-AS1
ATXN3

Clear

Reference Gene List

genome_protein-coding

Select Reference Set for Enrichment Analysis ⓘ Reset

粘贴100个MYC蛋白最频繁结合的基因组位置的gene symbol

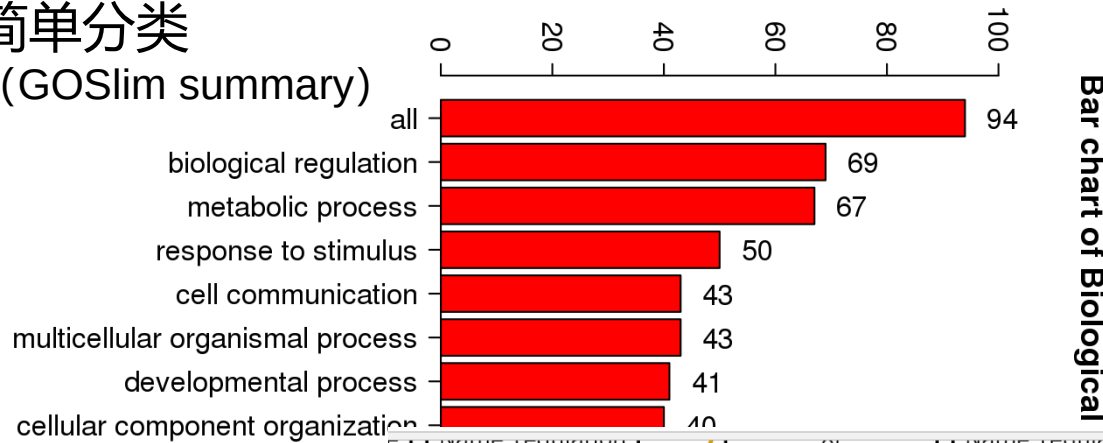
<http://www.webgestalt.org/>

另一个常用基因功能富集网站: DAVID, <https://david.ncifcrf.gov/>

WebGestalt 分析结果

简单分类

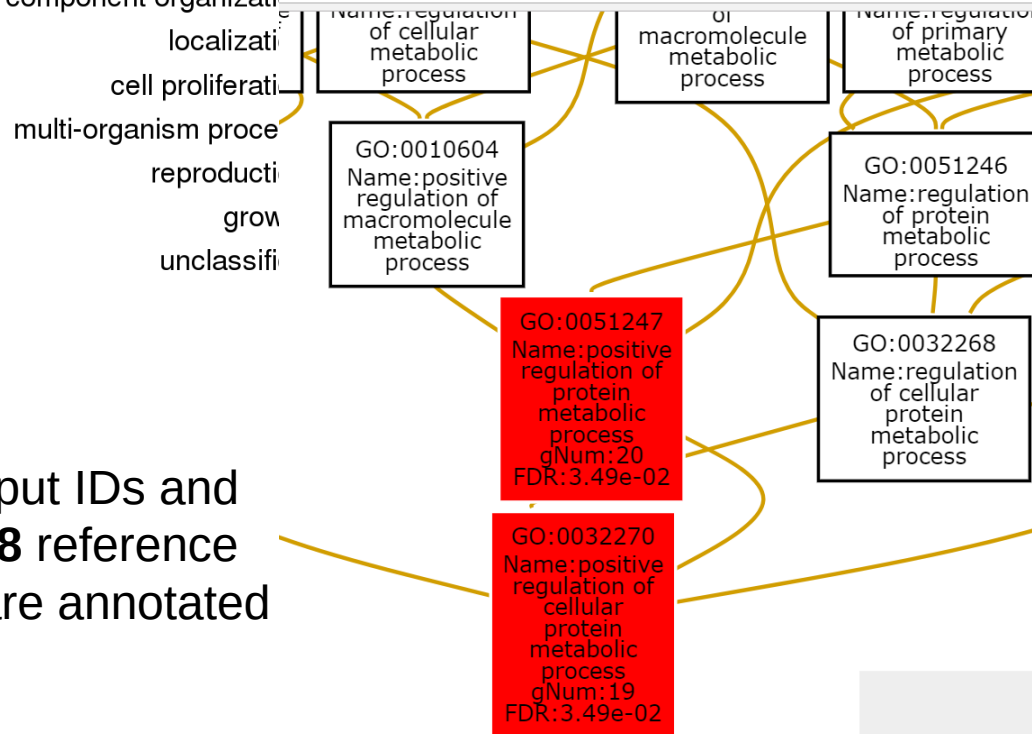
(GOSlim summary)



Enrichment Results

- C: GO category size
- O: observed genes
- E: Expected genes
- R: Enrichment ratio

82 input IDs and
16258 reference
IDs are annotated



ID:GO:0051247 Name:positive regulation of protein metabolic process

C=1485; O=20; E=7.03; R=2.84;
PValue=1.32e-05; FDR=3.49e-02

userid	Gene Symbol	Gene Name	Entrez Gene
ABI1	ABI1	abl interactor 1	10006
BCL2L11	BCL2L11	BCL2 like 11	10018
ANAPC10	ANAPC10	anaphase promoting complex subunit 10	10393
GLIPR2	GLIPR2	GLI pathogenesis related 2	152007
DAG1	DAG1	dystroglycan 1	1605

Pathway/KEGG富集分析

ID	Name	#Gene	FDR
hsa01521	EGFR tyrosine kinase inhibitor resistance - Homo sapiens (human)	3	7.32e-01
hsa05166	HTLV-I infection - Homo sapiens (human)	5	7.32e-01
hsa04215	Apoptosis - multiple species - Homo sapiens (human)	2	7.32e-01
hsa04914	Progesterone-mediated oocyte maturation - Homo sapiens (human)	3	7.32e-01
hsa04933	AGE-RAGE signaling pathway in diabetic complications - Homo sapiens (human)	3	7.32e-01
hsa05340	Primary immunodeficiency - Homo sapiens (human)	2	7.32e-01

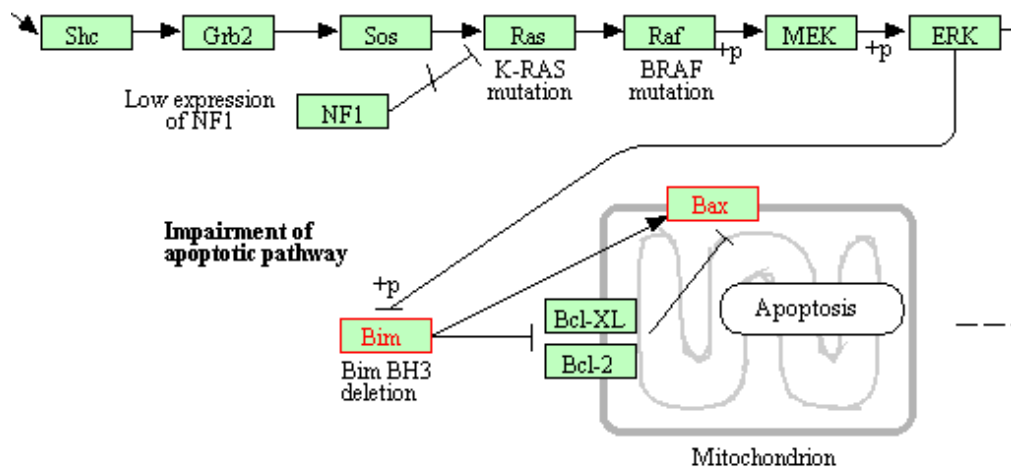
ID Search

[Download Table](#)

ID: hsa01521 **Name:** EGFR tyrosine kinase inhibitor resistance - Homo sapiens (human)

C=81; O=3; E=0.43; R=6.99; PValue=8.81e-03; FDR=7.32e-01

userid	Gene Symbol	Gene Name	Entrez Gene
BCL2L11	BCL2L11	BCL2 like 11	10018
BAX	BAX	BCL2 associated X, apoptosis regulator	581
VEGFA	VEGFA	vascular endothelial growth factor A	7422



生信思路:

- 关注预期和意料之外的富集通路
- 关注输入基因列表和通路的交集基因（通路图中标为红色）、它们在通路中的上下游关系（富集分析中没有用到网络结构信息）

法则 4：思考备选解释

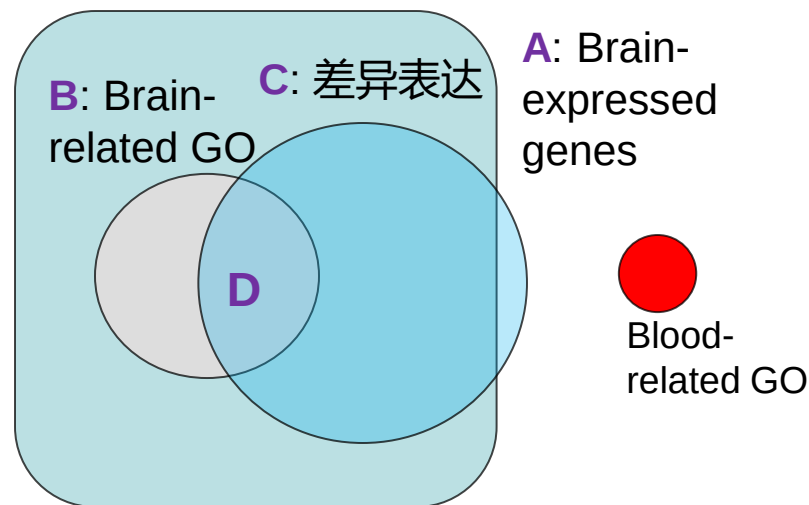
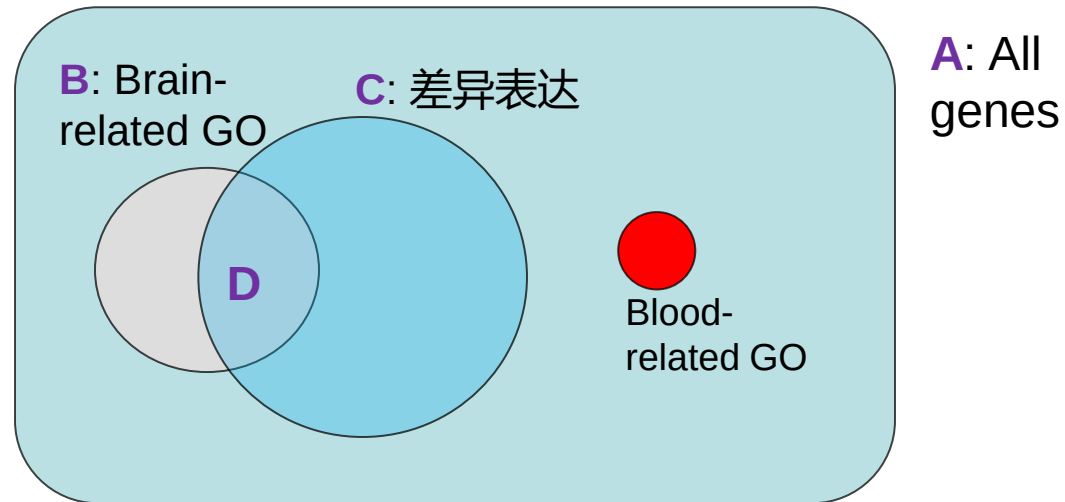
举例：富集分析的背景基因选择

分析功能基因组的数据时，我们经常会得到一个人类基因列表，比如在正常和癌症细胞里有差异表达的基因。一个一个看这些基因的已知功能的描述太繁琐，也不容易看出规律。一个常用的方法是Gene Ontology的**富集分析**（比如<https://david.ncifcrf.gov/>）。许多基因的功能可以用一些规范的条目（GO terms）表来描述，这个分析方法就能找出我的基因列表中富集的GO条目。

下面两种分析方法你认为那个更合理呢：

- 1) 直接提交我的基因列表，这个网站会用所有人类基因作为比较的背景，使用超几何分布来找出显著富集的GO条目。这些功能就可能是和我在研究的癌症相关的。许多论文不是这么做的吗？
- 2) 假如我的差异表达数据是关于大脑的疾病，**因为大部分有差异表达的基因也在大脑细胞中表达**，即使我的基因列表和这种疾病没有关系（比如随机从在脑细胞中表达的基因中选取一个子集），它也会富集一些GO条目（比如和脑功能相关的）。所以，要在富集分析的网站上同时提交一个脑细胞中表达的基因列表作为“Reference gene list”来校正这种影响。

富集分析中背景基因的影响



分析思路

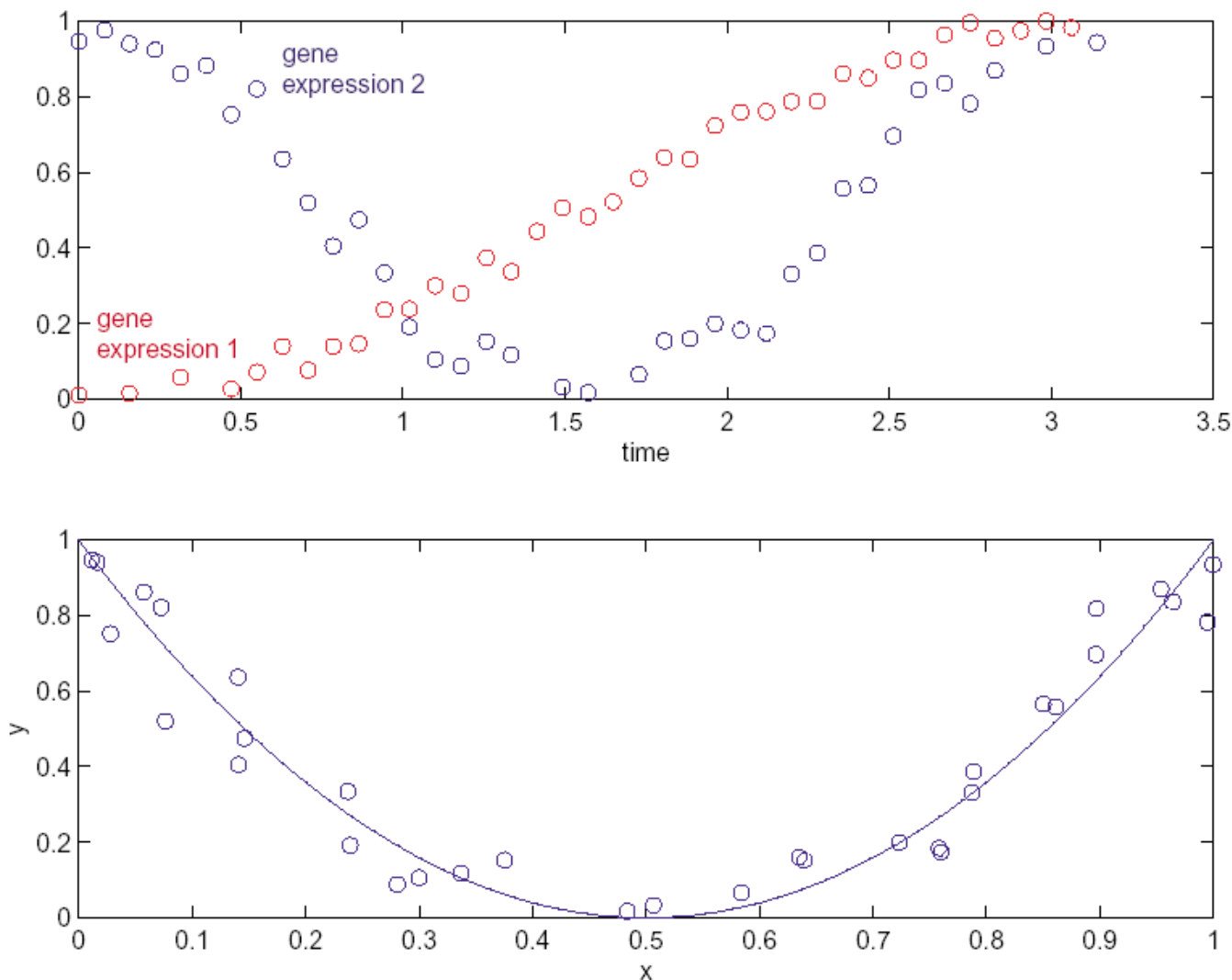
(刘玉婷) :
分别看上调和
下调的基因中
富集的通路,
是否有异同?

- **A:** 列联表分析考虑的所有基因
- **C:** 上页中差异表达的基因, 大都是大脑表达基因
- 富集分析比较的比例: **D/C vs. B/A**

课后练习

1. 阅读文献《Immunity 18 Lineage-Determining Transcription Factor TCF-1 Initiates the Epigenetic Identity of T Cells》，找出哪些图用到了基因富集分析？显著性富集提示了什么样的生物功能？
2. 查看R语言相关函数（“R functions for count data”页）的参数和例子（例如用[fisher.test](#)）。使用相关函数检验“[比例的比较：TCF-1](#)”例子中的频率差异是否显著？
3. 上网搜索(如[www.bing.com](#))，发现使用R语言做基因功能富集、GSEA分析的软件包和教程。试用MYC 靶基因列表（[MYC binding.xlsx](#)）做富集分析。

课后练习4：How to detect nonlinear relationship?



提示：将X、Y数值转换为高、中、低三类，做3x3列联表